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(54) **INTEGRATED CIRCUIT WITH A FIN AND GATE STRUCTURE AND METHOD MAKING THE SAME**

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**H10D 84/03** (2025.01)

**H10D 84/83** (2025.01)

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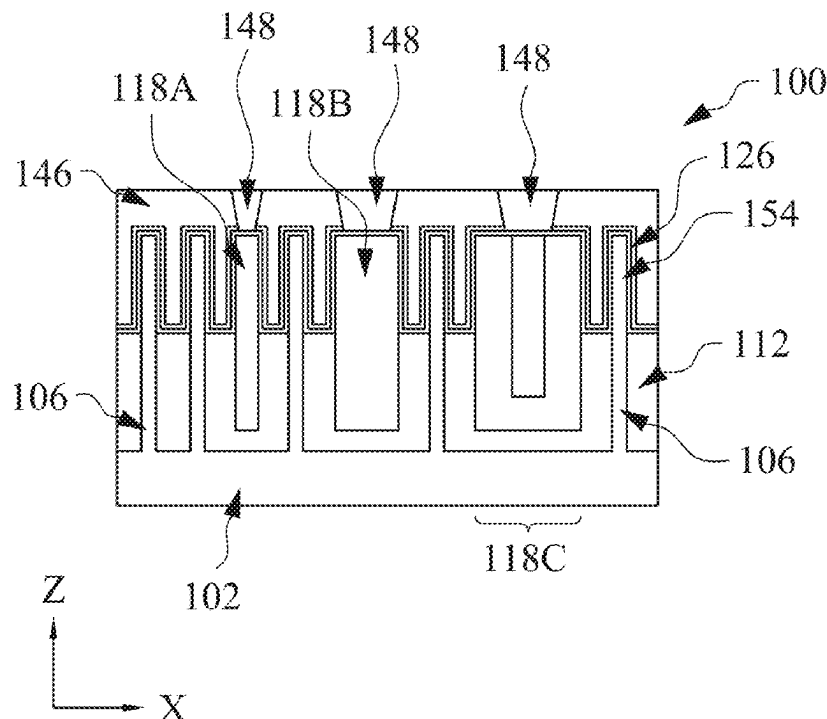
(52) **U.S. Cl.**

**CPC ... H10D 30/0243** (2025.01); **H01L 21/76229** (2013.01); **H10D 62/832** (2025.01); **H10D 64/017** (2025.01); **H10D 64/021** (2025.01); **H10D 84/0135** (2025.01); **H10D 84/0151** (2025.01); **H10D 84/0158** (2025.01); **H10D 84/0193** (2025.01); **H10D 84/038** (2025.01); **H10D 84/834** (2025.01); **H10D 84/853** (2025.01); **H10D 30/6219** (2025.01); **H10D 30/797** (2025.01); **H10D 62/822** (2025.01)

(57)

**ABSTRACT**

The present disclosure provides a semiconductor structure. The semiconductor structure includes device fins formed on a substrate; fill fins formed on the substrate and disposed among the device fins; and gate stacks formed on the device fins and the fill fins. The fill fins include a first dielectric material layer and a second dielectric material layer deposited on the first dielectric material layer. The first and second dielectric material layers are different from each other in composition.



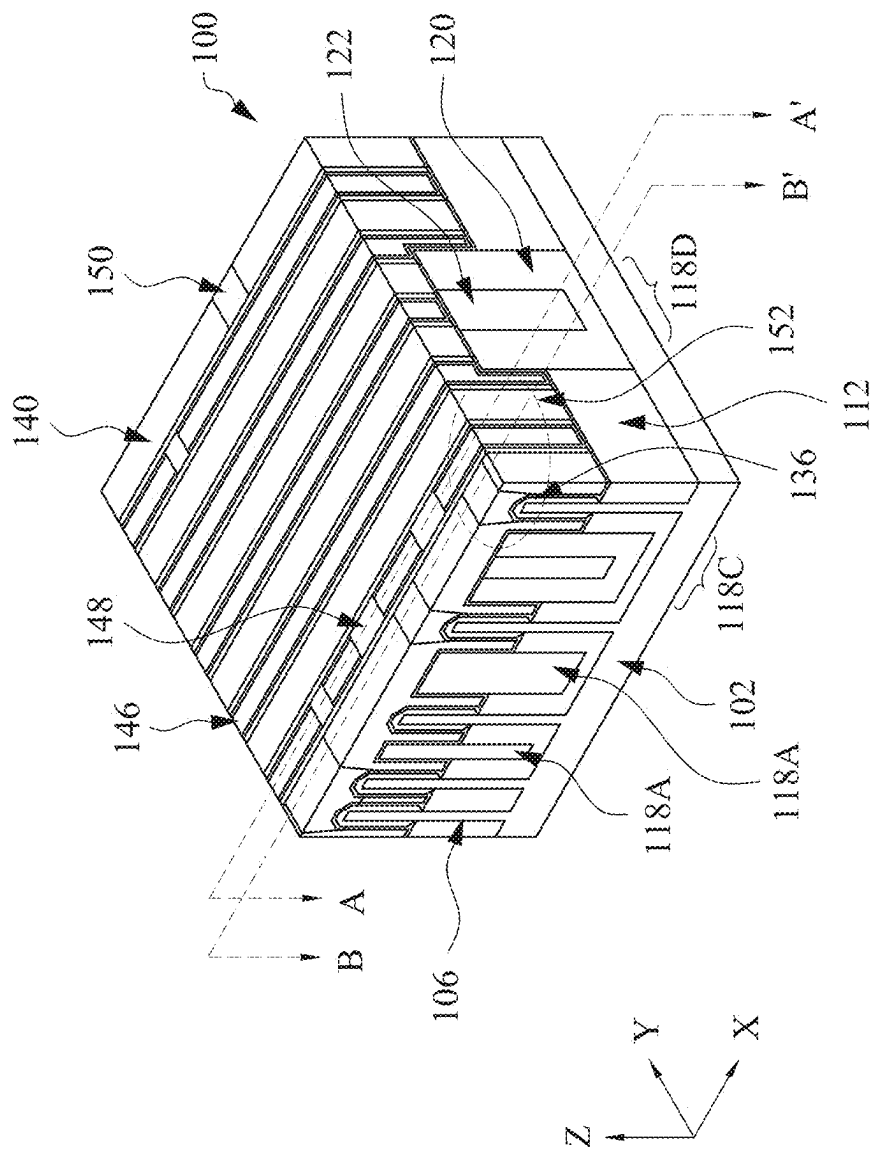


FIG. 1A

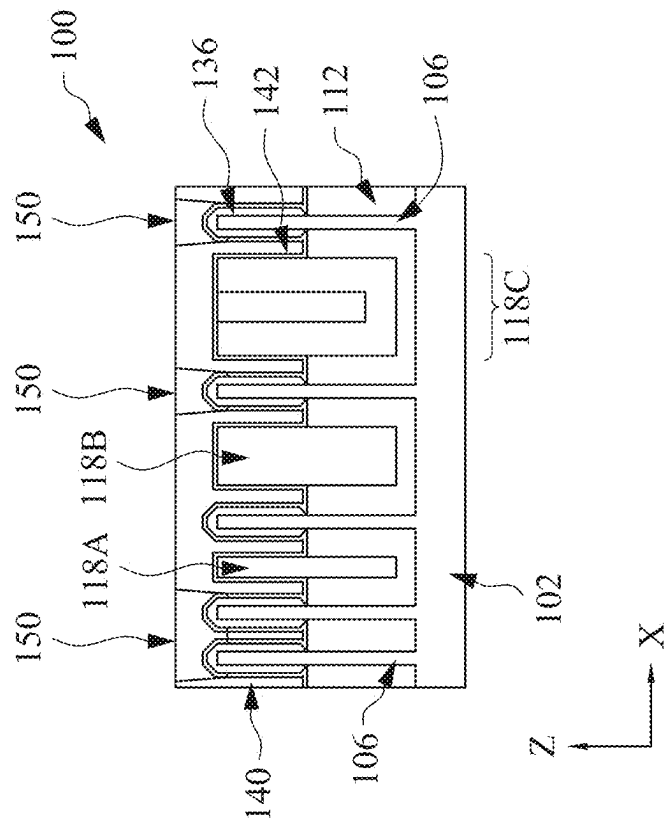


FIG. 1C

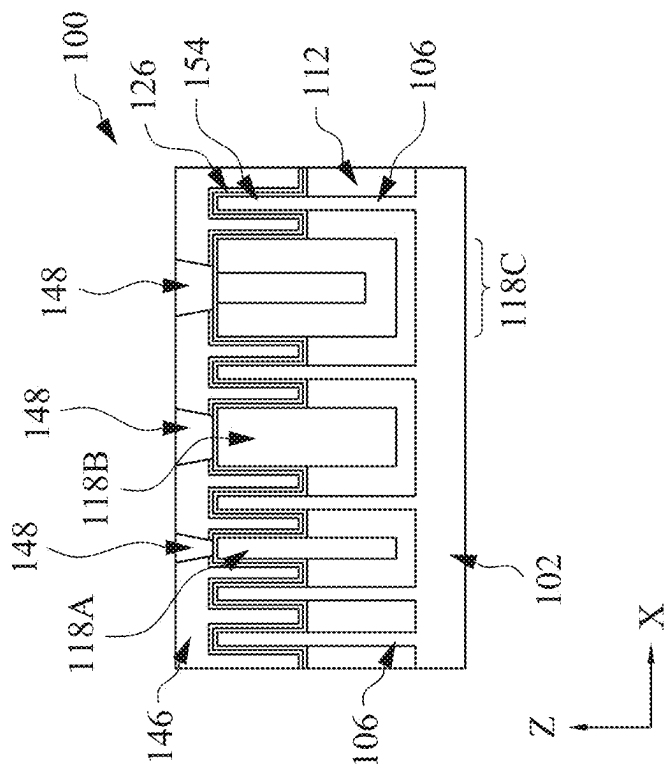


FIG. 1B

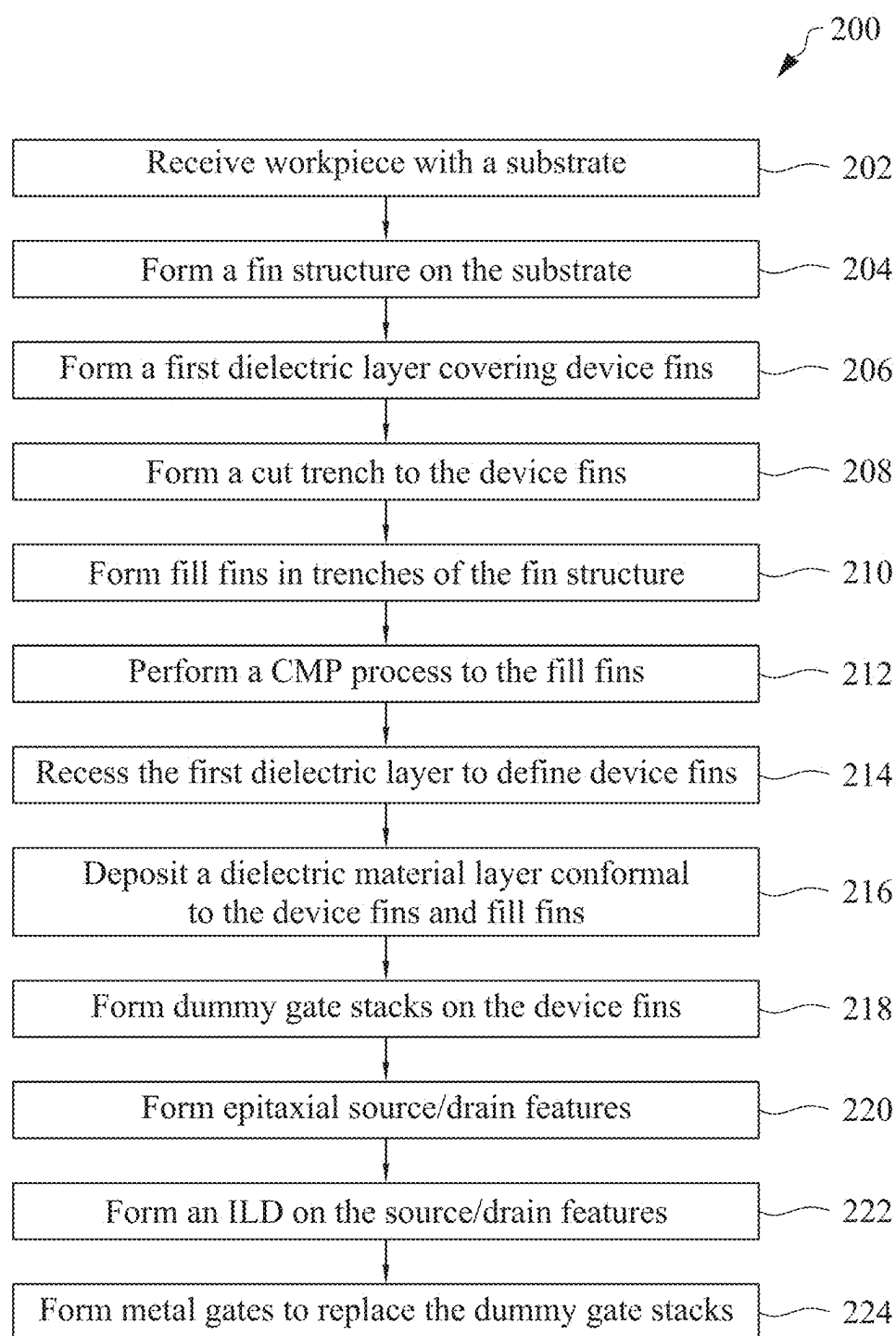


FIG. 2

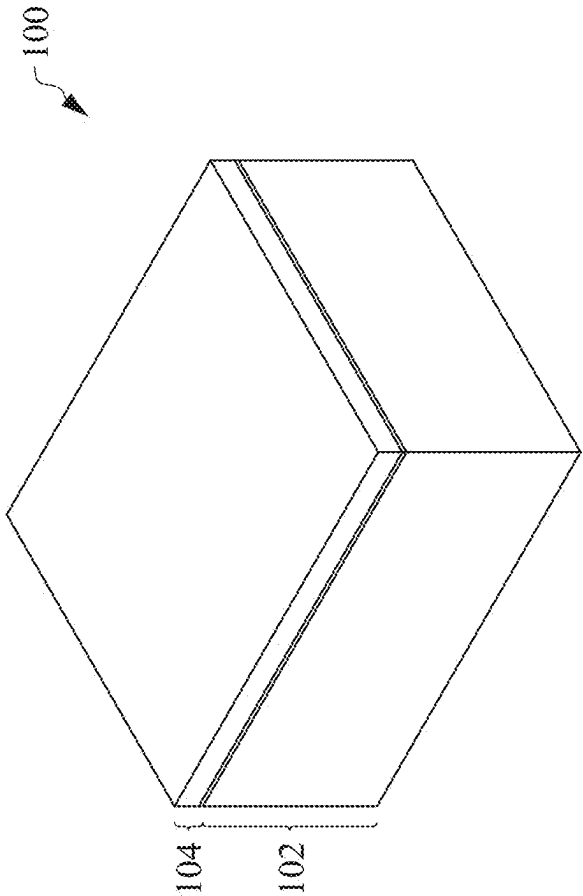


FIG. 3

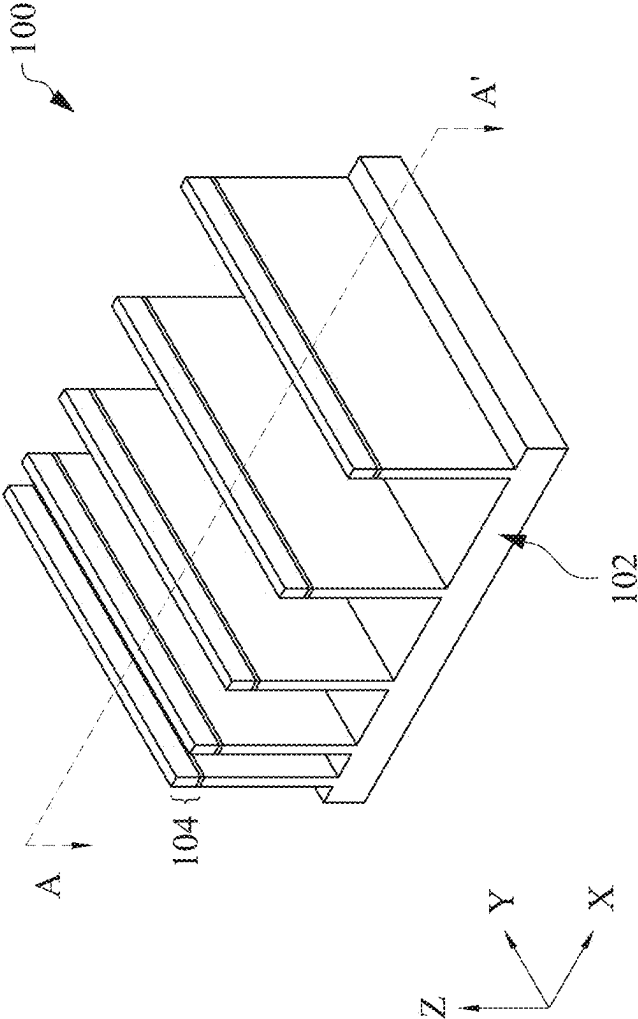


FIG. 4A

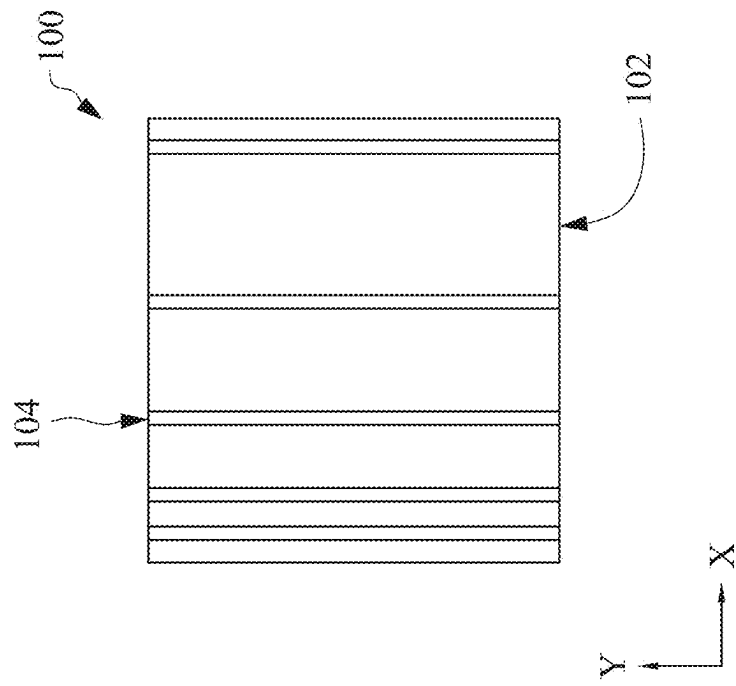


FIG. 4B

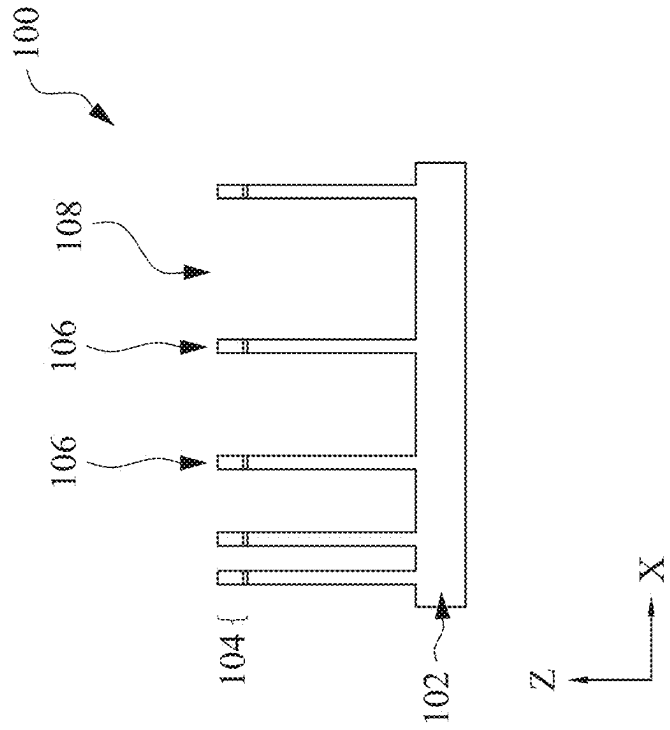


FIG. 4C

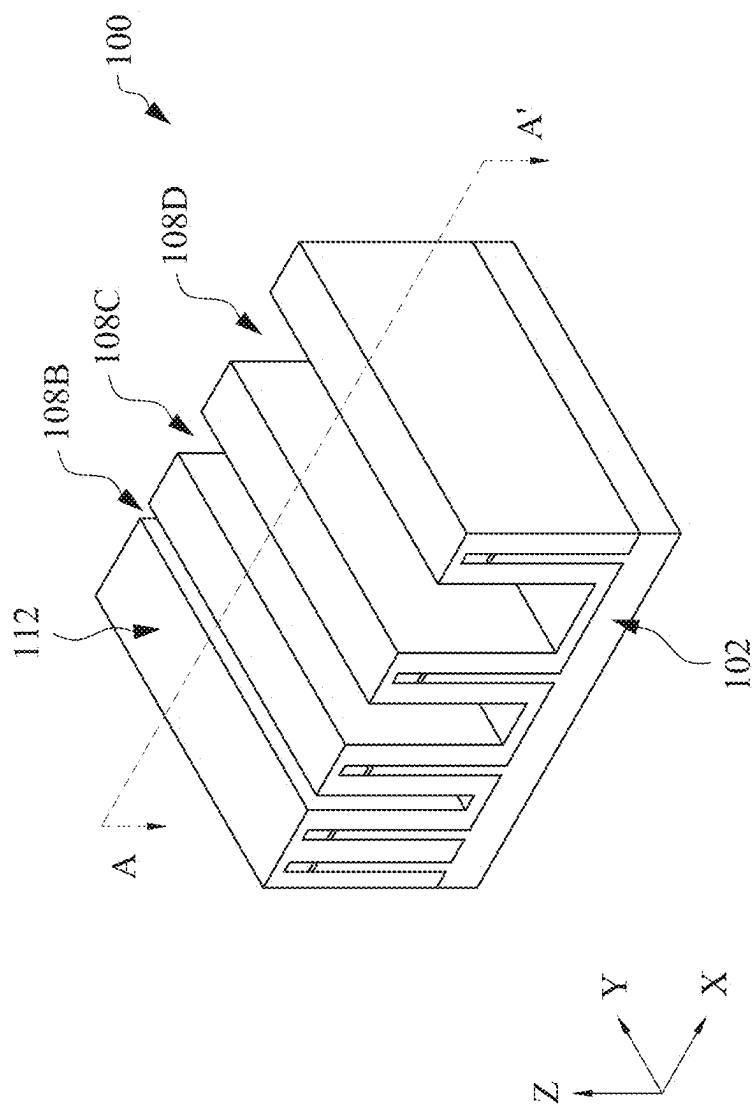


FIG. 5A



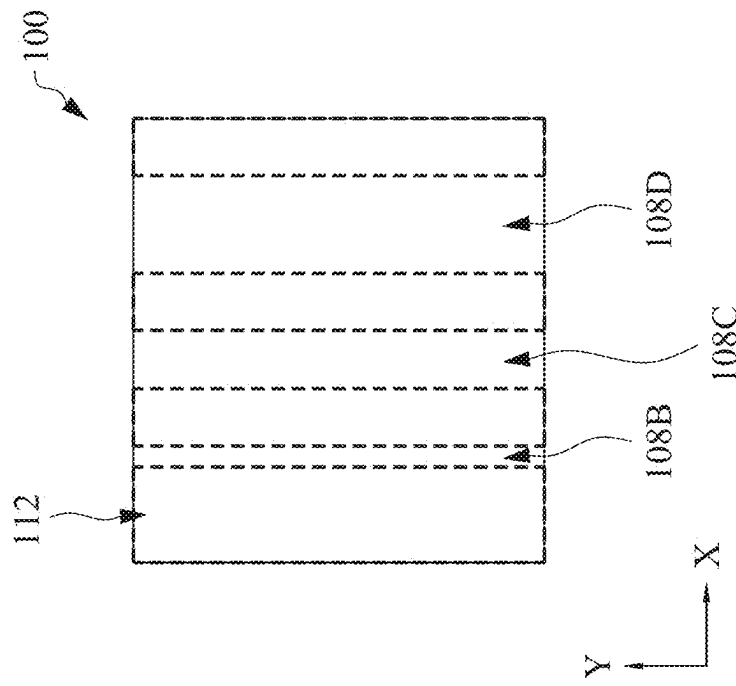


FIG. 5B

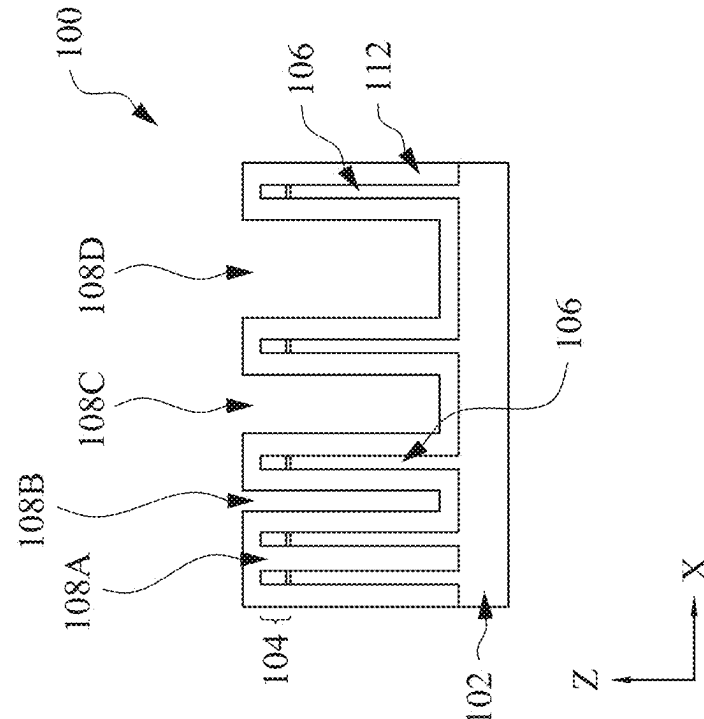


FIG. 5C

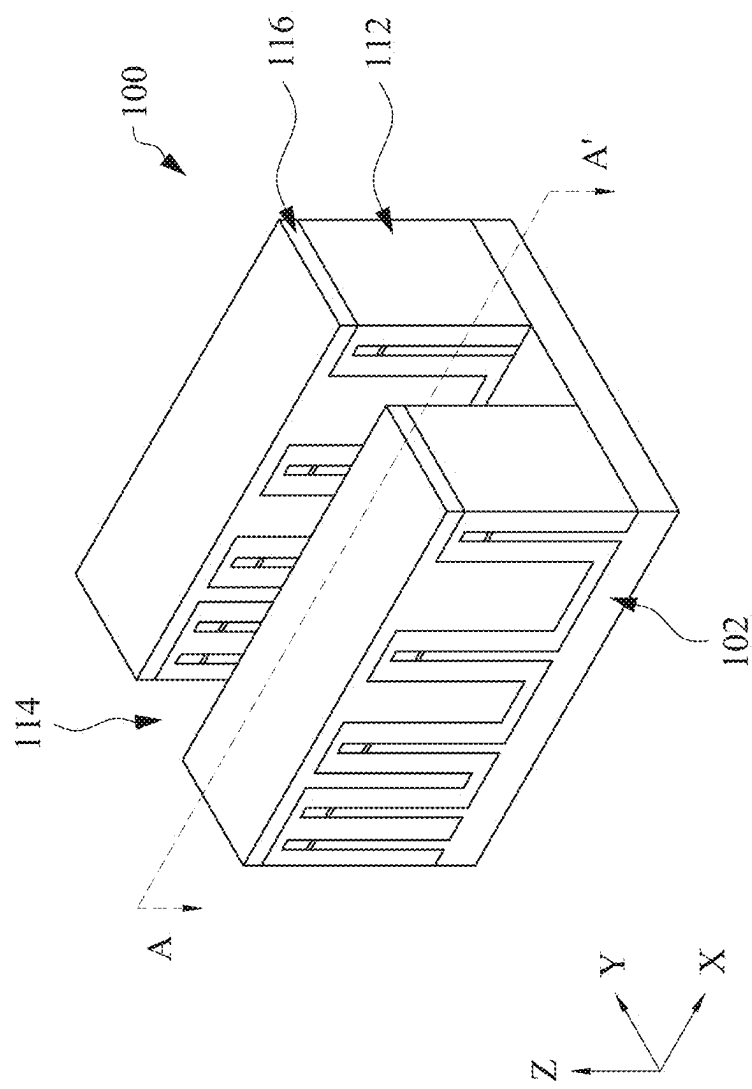


FIG. 6A

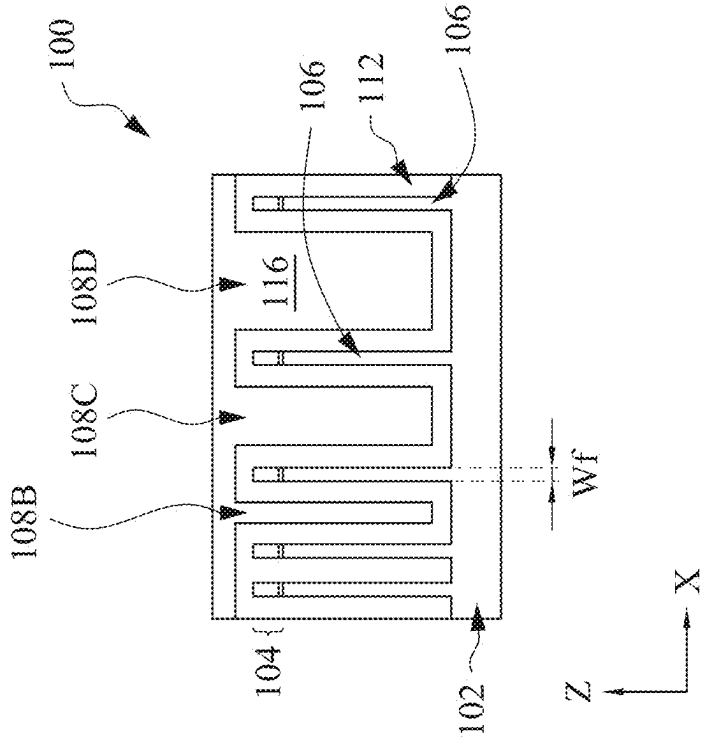


FIG. 6C

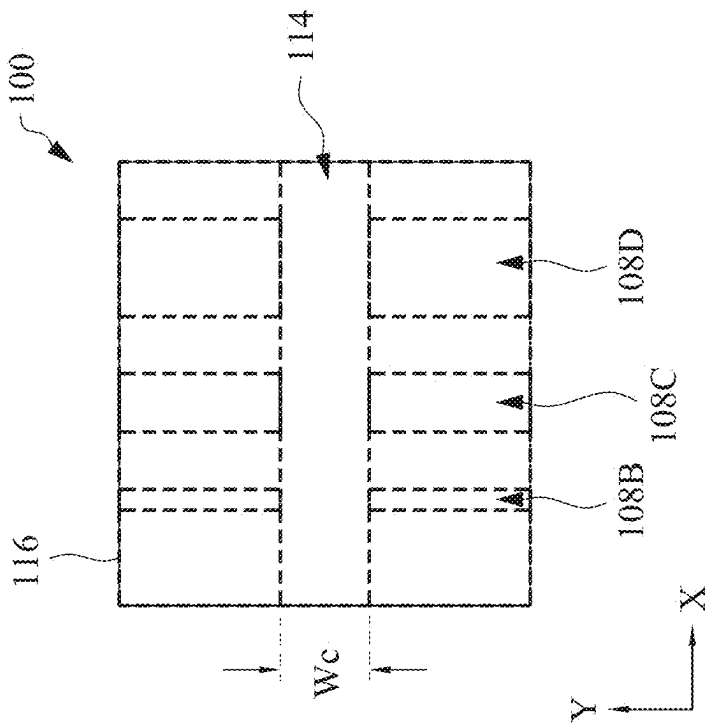


FIG. 6B

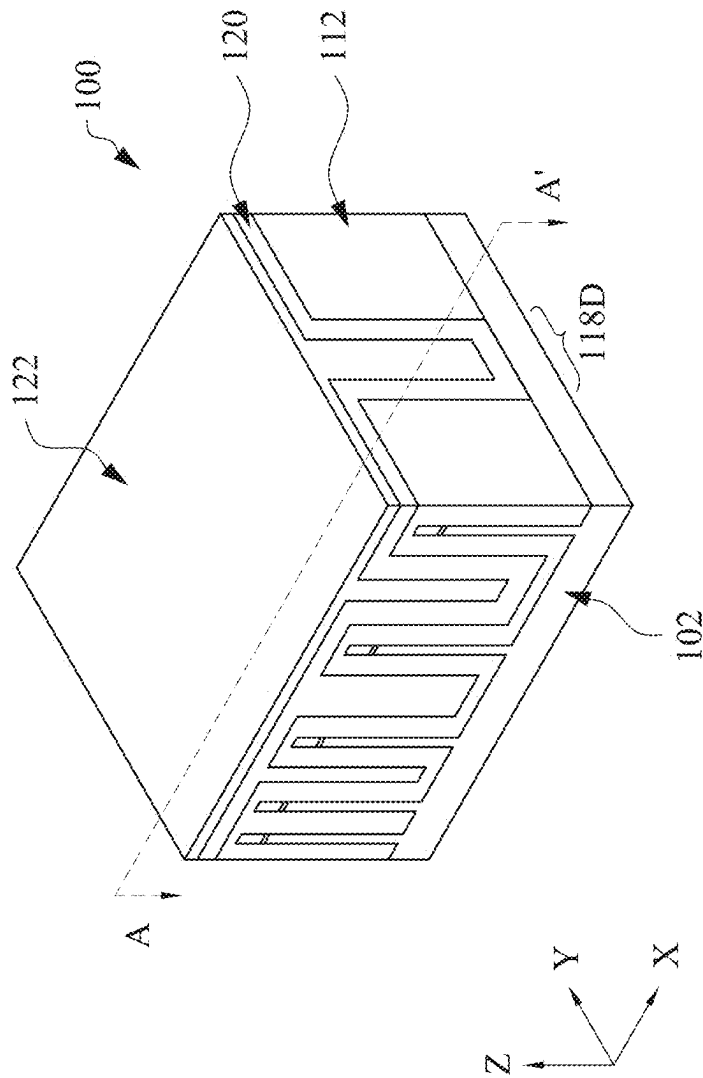


FIG. 7A

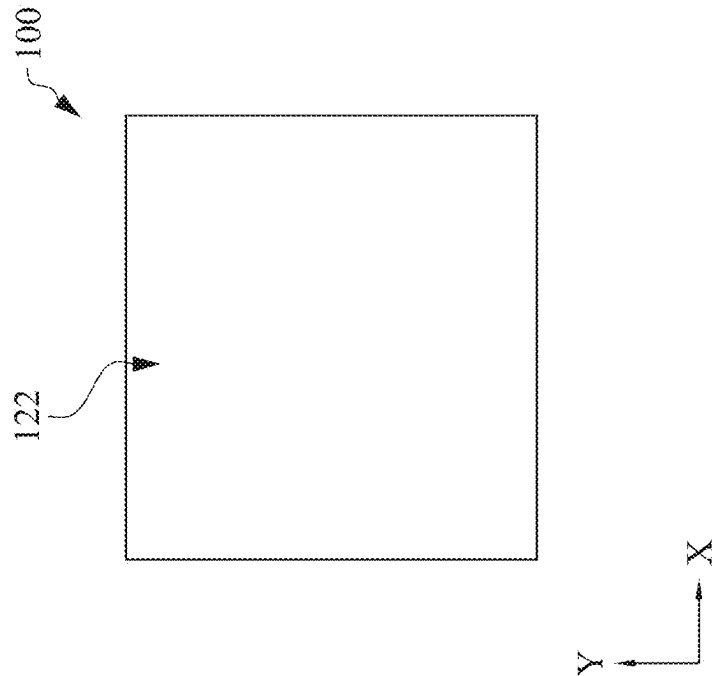


FIG. 7B

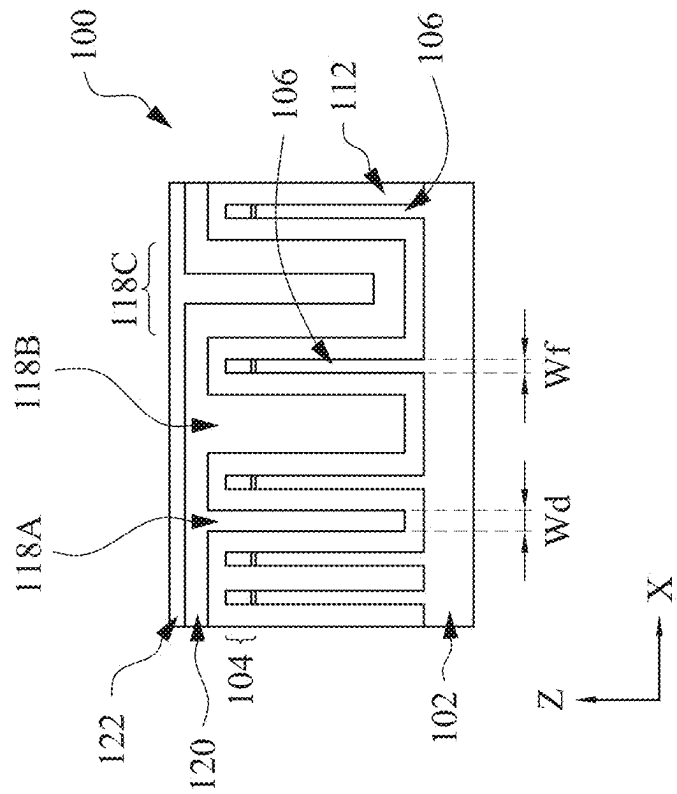


FIG. 7C

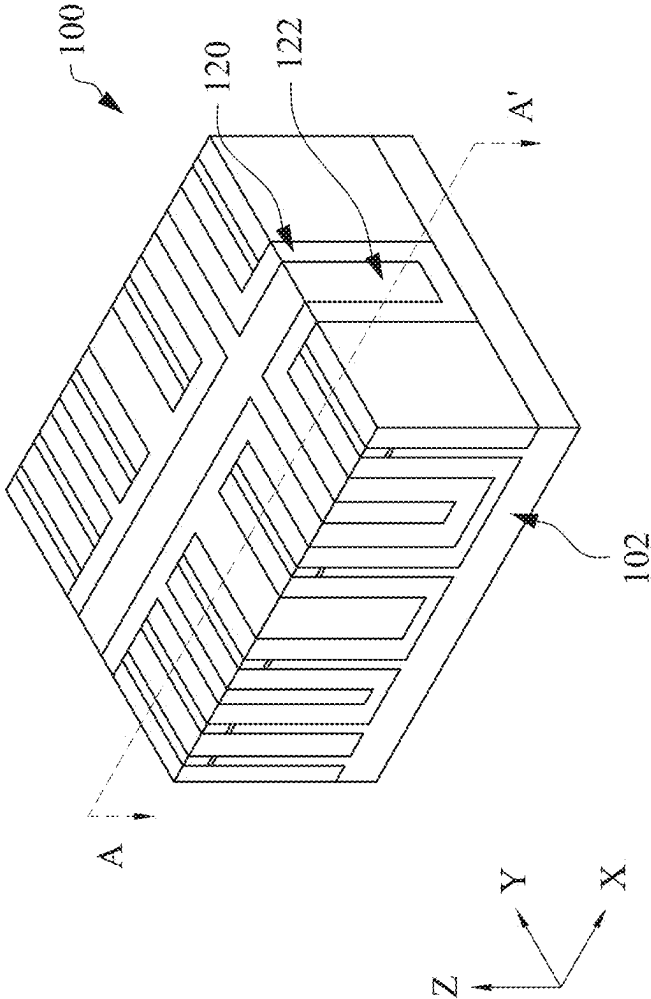


FIG. 8A

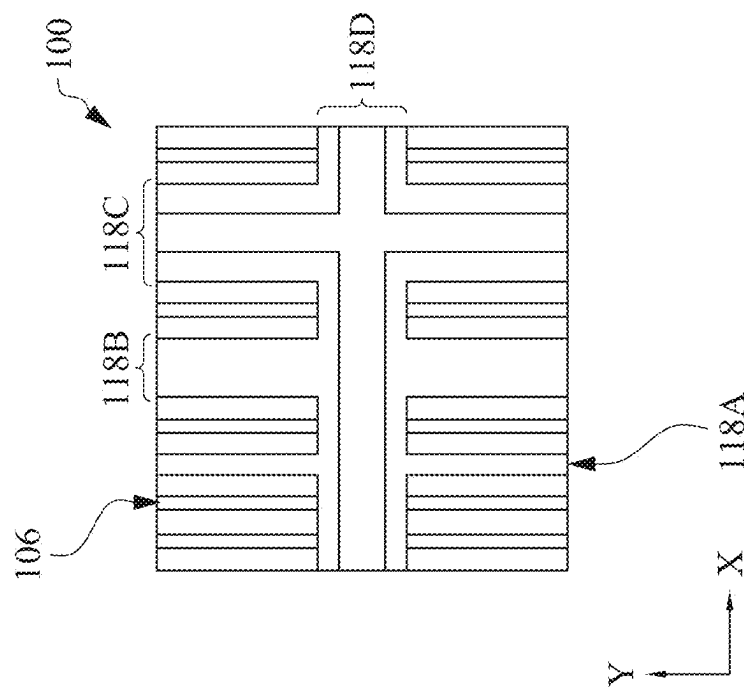


FIG. 8B

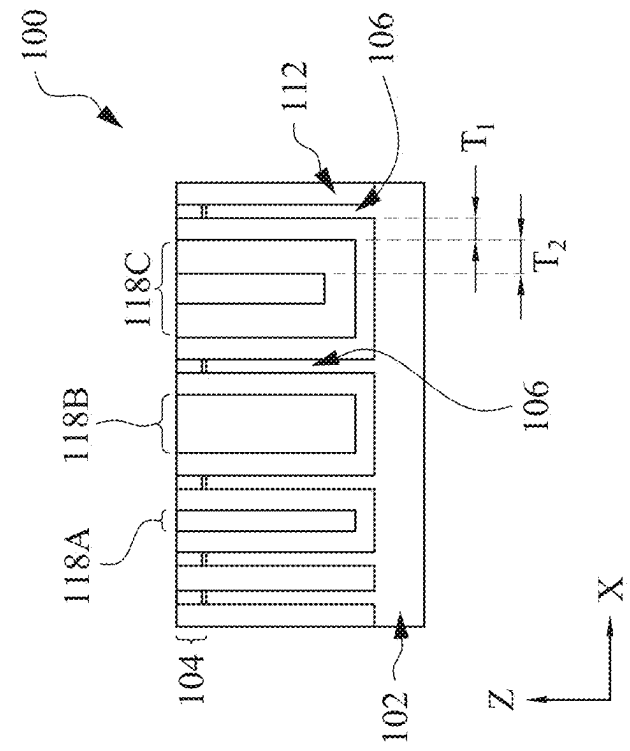


FIG. 8C

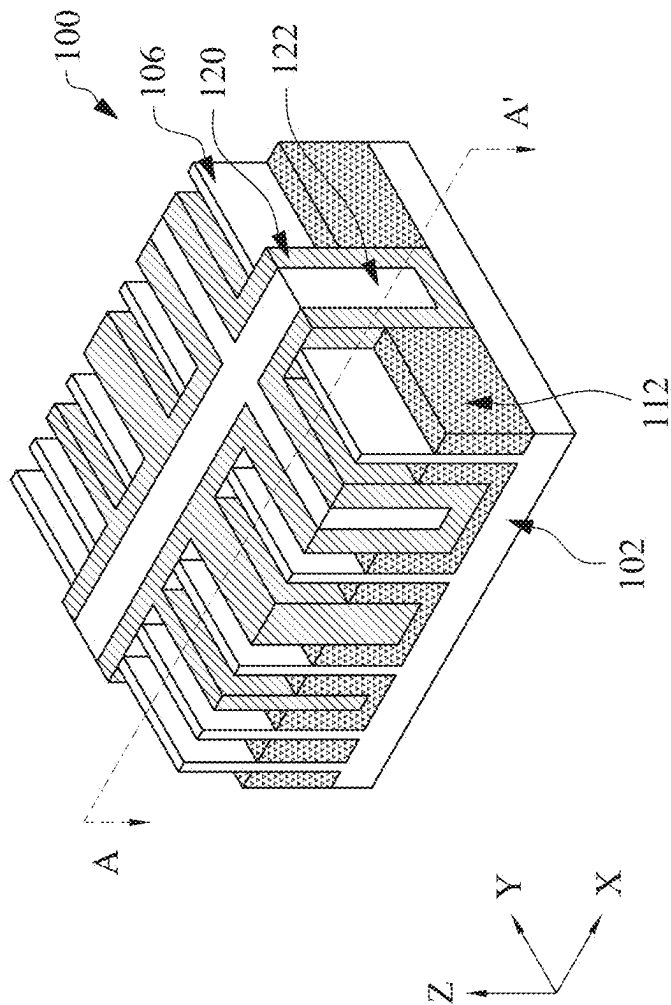


FIG. 9A



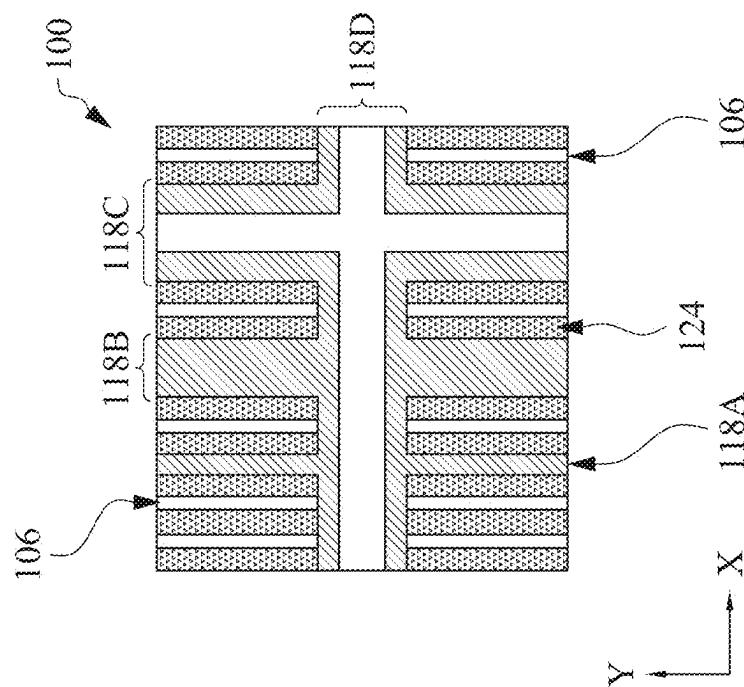


FIG. 9B

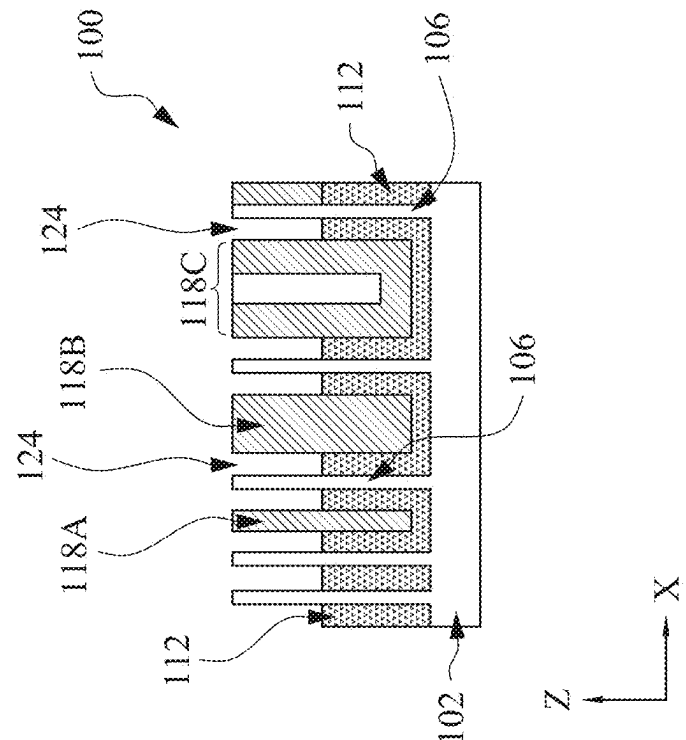


FIG. 9C

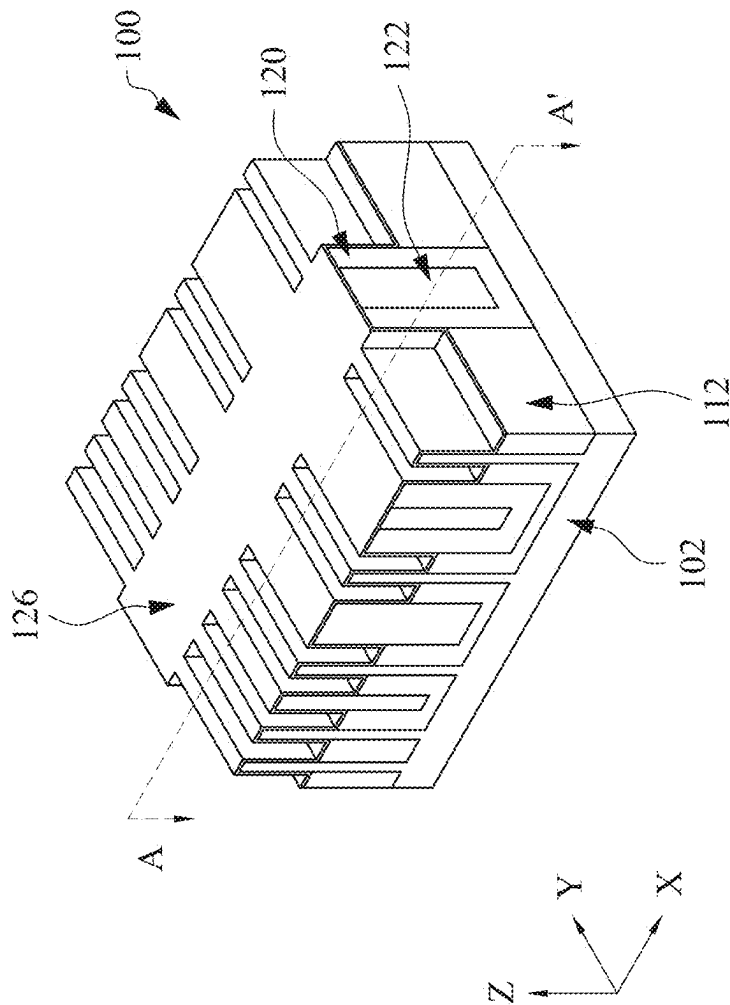


FIG. 10A

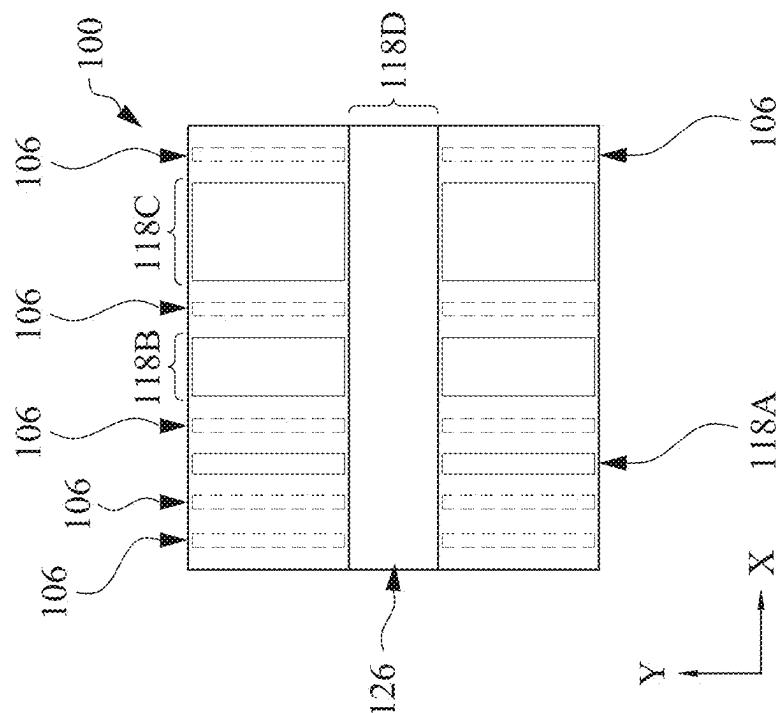


FIG. 10B

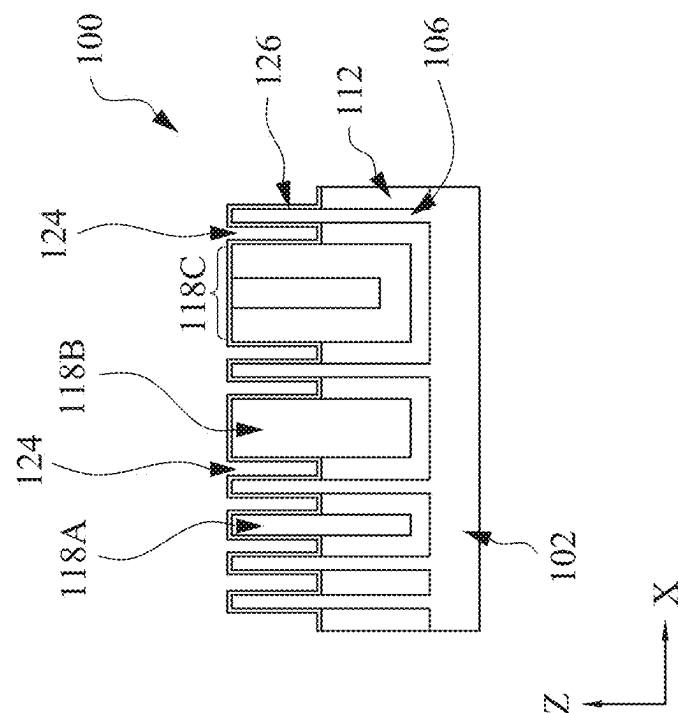


FIG. 10C

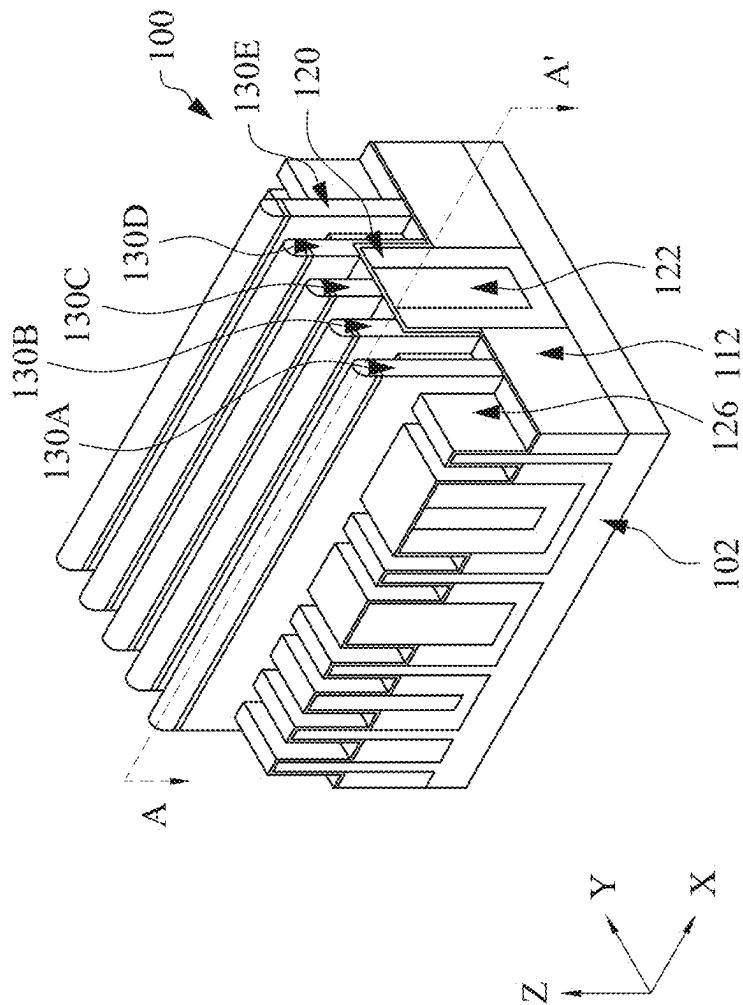


FIG. 11A

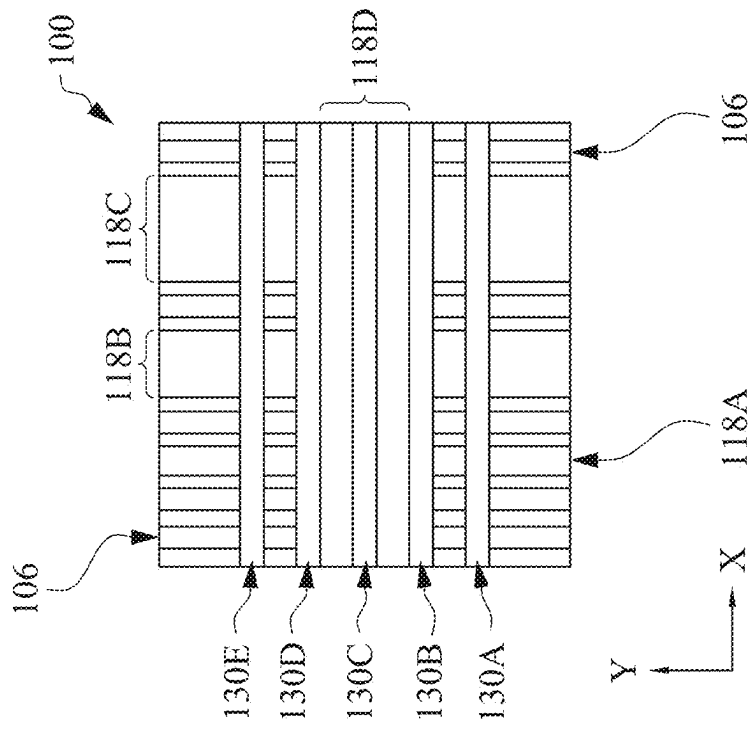


FIG. 11B

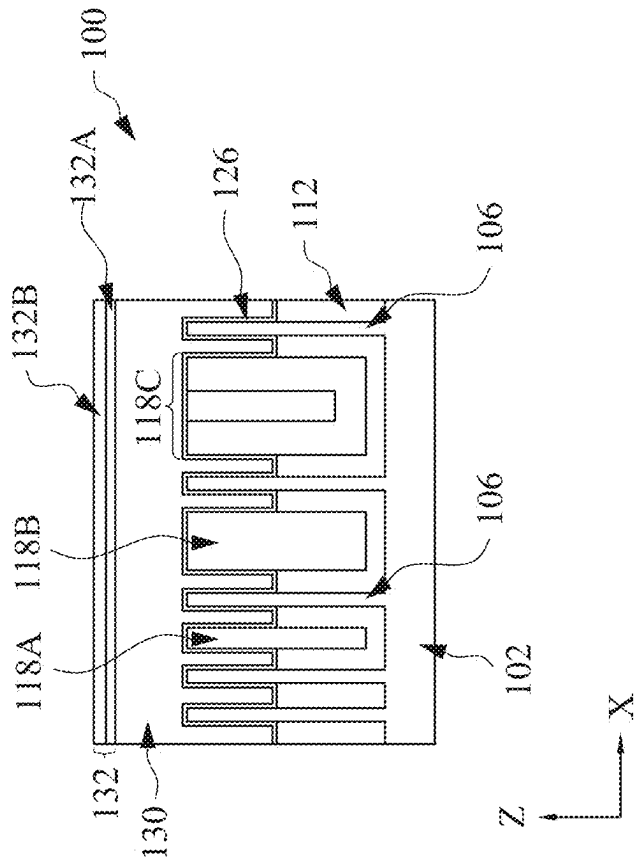


FIG. 11C

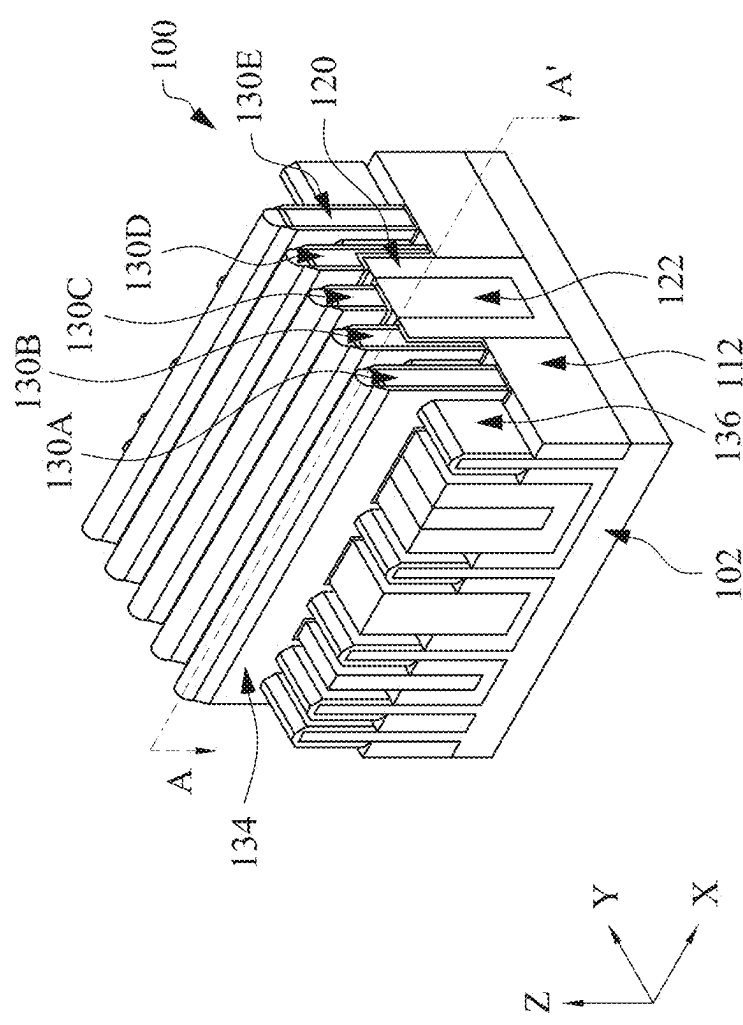


FIG. 12A

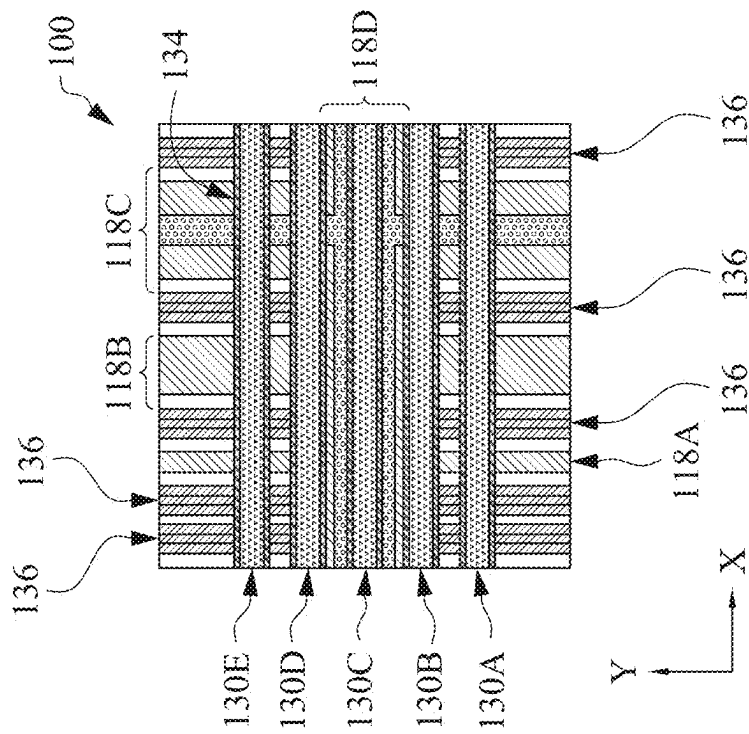


FIG. 12B

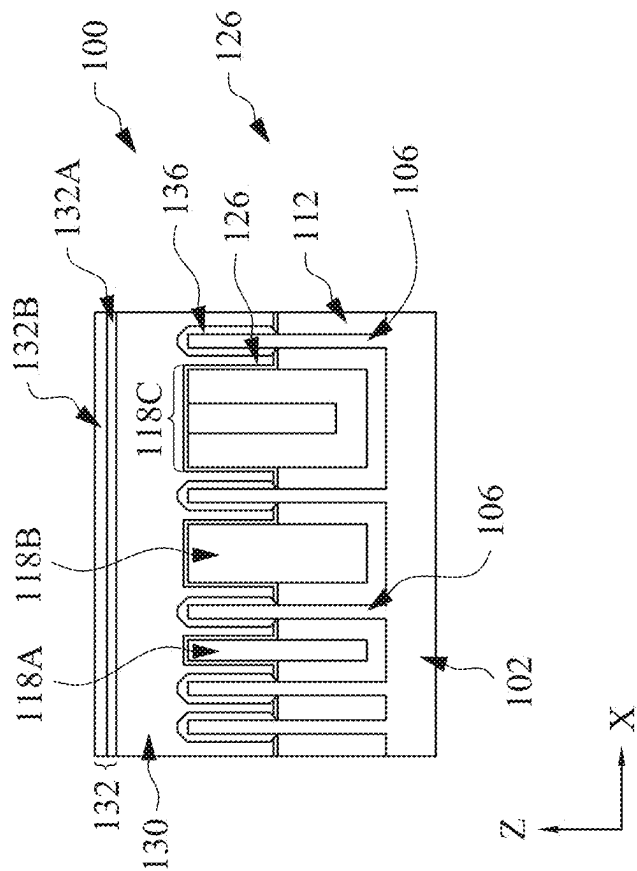


FIG. 12C

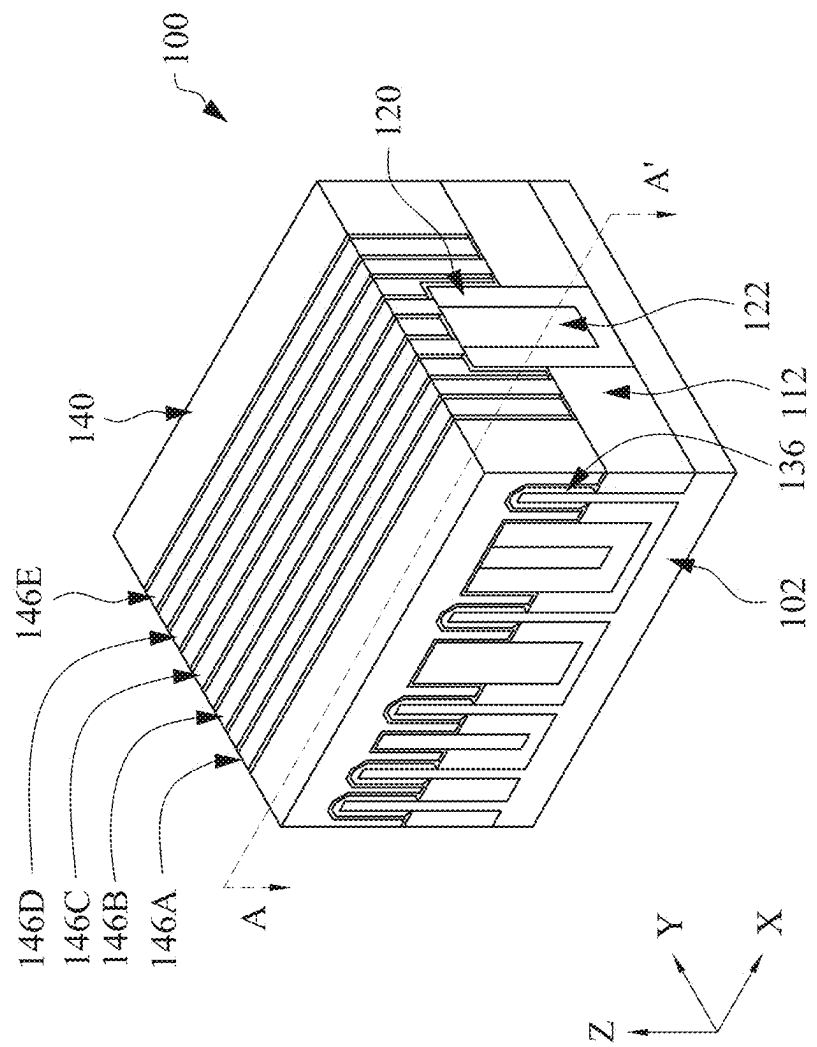


FIG. 13A



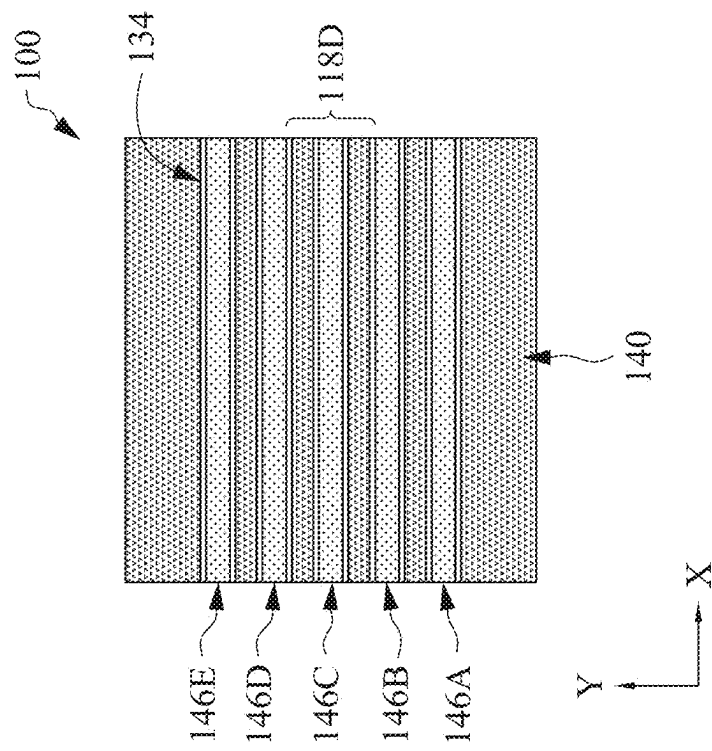


FIG. 13B

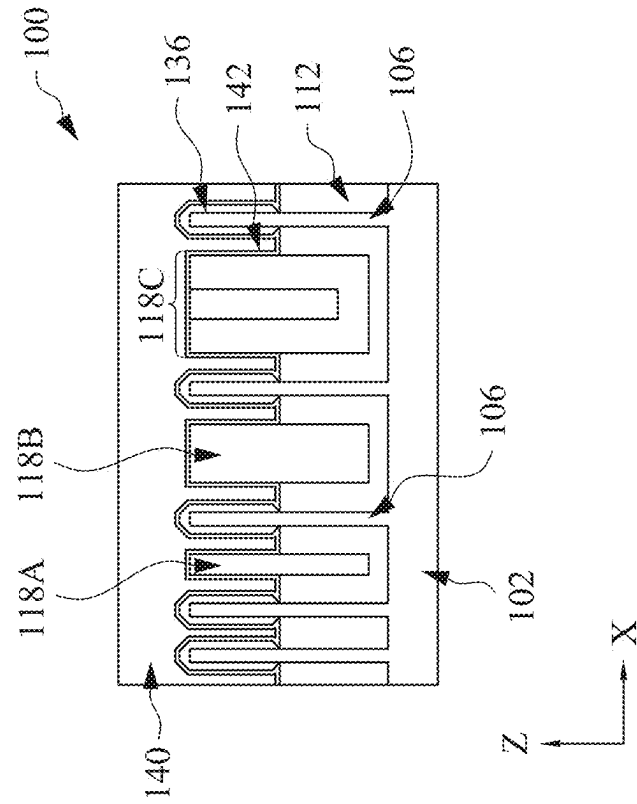


FIG. 13C

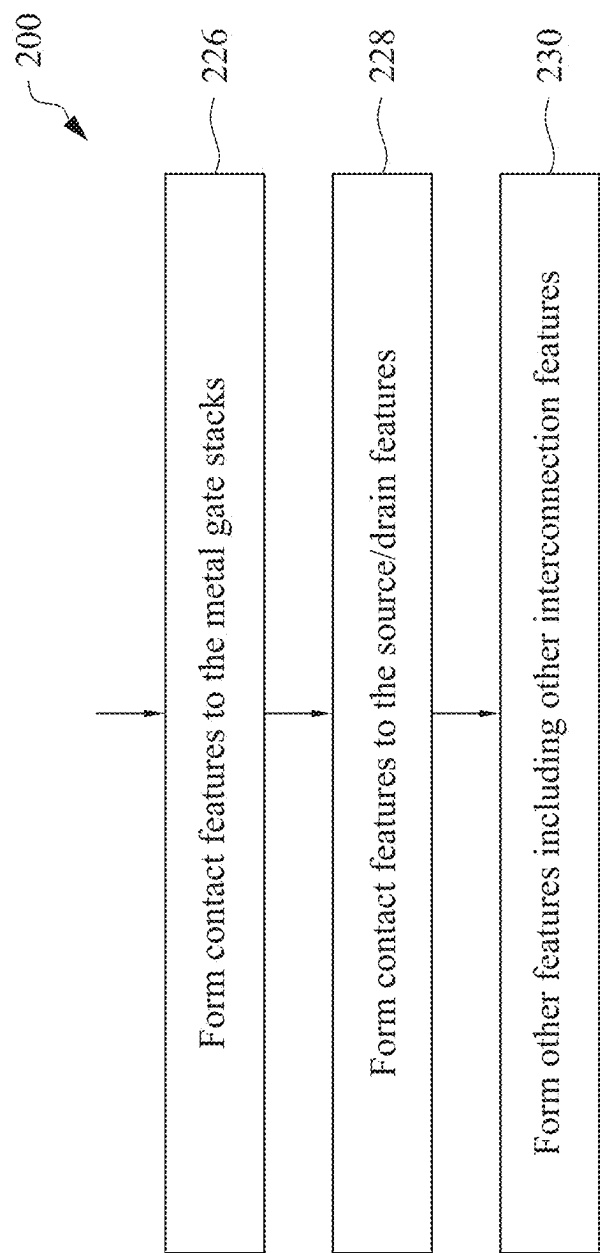


FIG. 14

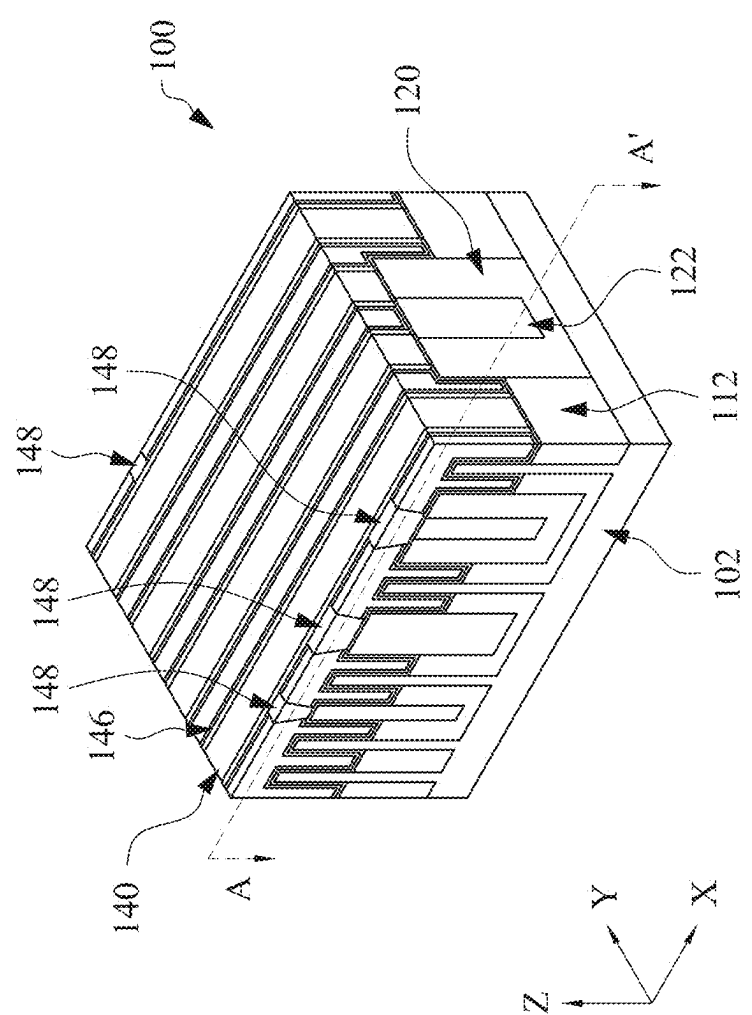


FIG. 15A

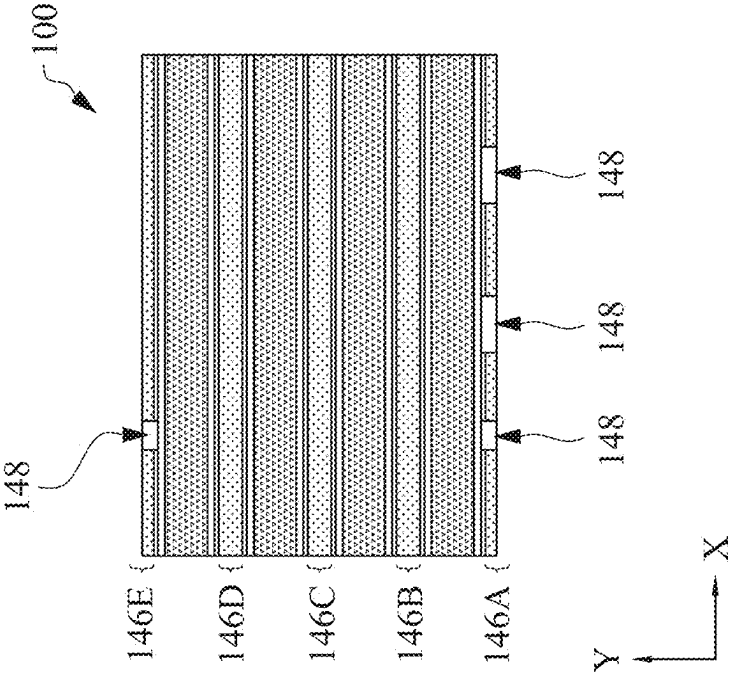


FIG. 15B

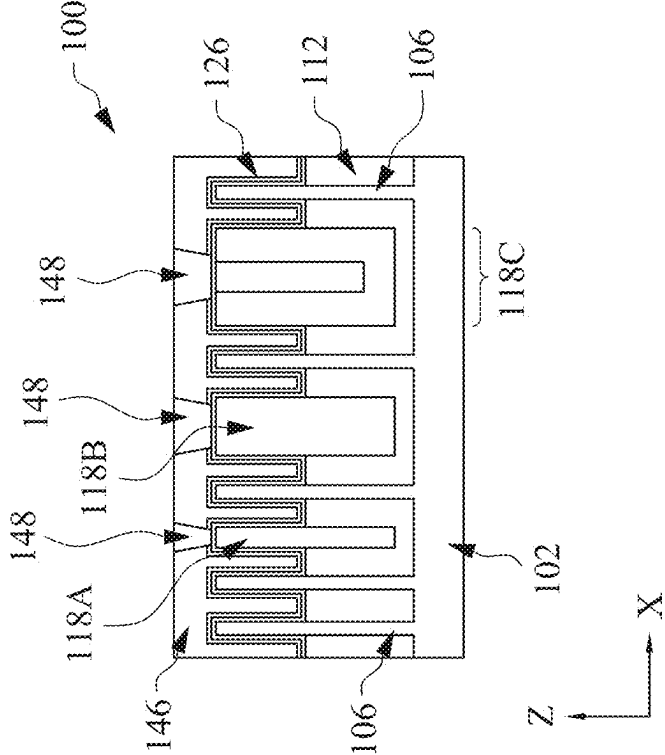


FIG. 15C

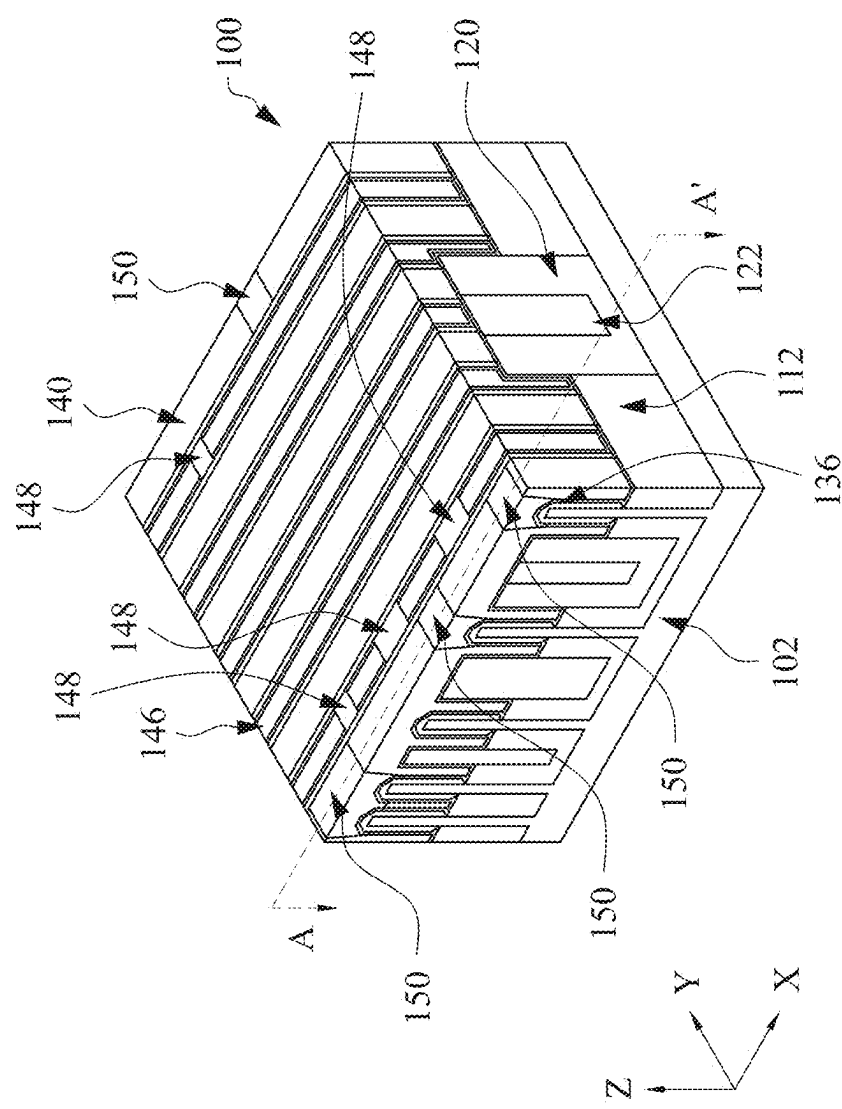


FIG. 16A

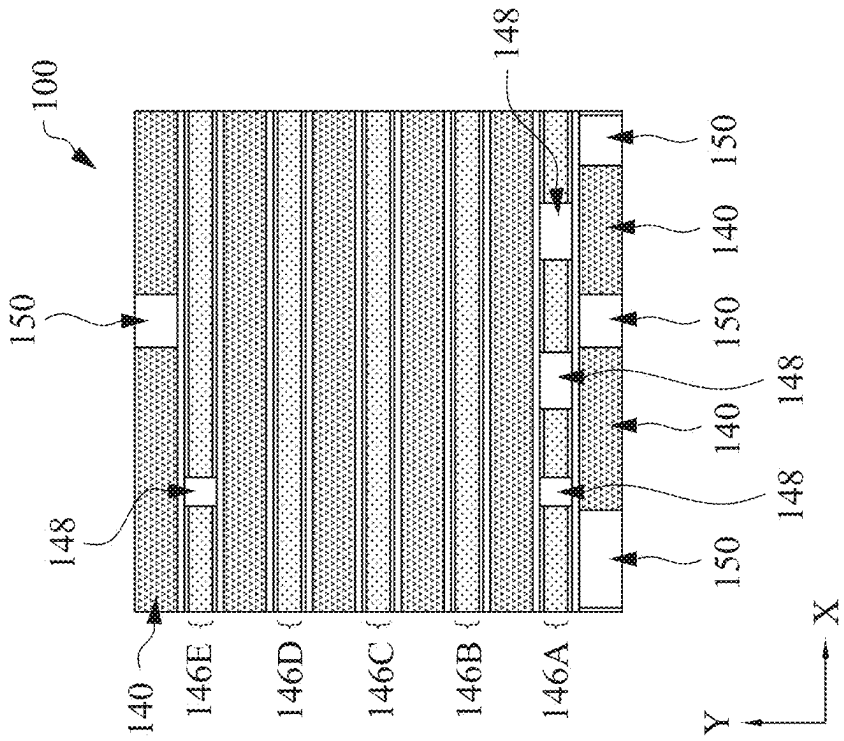


FIG. 16B

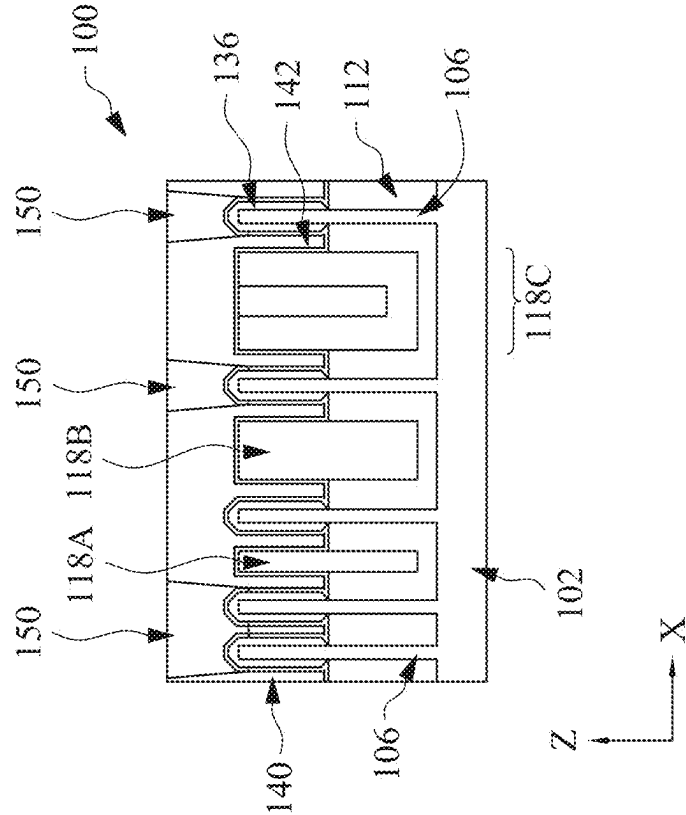


FIG. 16C

# INTEGRATED CIRCUIT WITH A FIN AND GATE STRUCTURE AND METHOD MAKING THE SAME

## PRIORITY

**[0001]** This application is a continuation of U.S. patent application Ser. No. 18/635,347, filed Apr. 15, 2024, which is a continuation of U.S. patent application Ser. No. 17/328,389, filed May 24, 2021 (now U.S. Pat. No. 11,961,900), which is a continuation of U.S. patent application Ser. No. 16/730,192, filed Dec. 30, 2019 (now U.S. Pat. No. 11,018,246), which is a continuation of U.S. patent application Ser. No. 15/864,525, filed Jan. 8, 2018 (now U.S. Pat. No. 10,529,833), which further claims the benefit of U.S. Provisional Application 62/551,044 entitled “Integrated Circuit With A Gate Structure and Method Making The Same,” filed Aug. 28, 2017, the entire disclosures of which are incorporated herein by reference.

## BACKGROUND

**[0002]** The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs and, for these advances to be realized. Similar developments in IC processing and manufacturing are needed. For example, a three-dimensional transistor, such as a fin-like field-effect transistor (FinFET), has been introduced to replace a planar transistor. A FinFET can be thought of as a typical planar device extruded into the gate. A typical FinFET is fabricated with a thin “fin” (or fin structure) extending up from a substrate. The channel of the FET is formed in this vertical fin, and a gate is provided over (e.g., wrapping around) the channel region of the fin. Wrapping the gate around the fin increases the contact area between the channel region and the gate and allows the gate to control the channel from multiple sides. This can be leveraged in a number of ways, and in some applications, FinFETs provide reduced short channel effects, reduced leakage, and higher current flow. In other words, they may be faster, smaller, and more efficient than planar devices.

**[0003]** However, because of the complexity inherent in FinFETs and other nonplanar devices, and further because of the high pattern density in the advanced technology nodes, a number of techniques used in manufacturing planar transistors are not well suited to fabricating nonplanar devices. As merely one example, conventional techniques for forming gate stacks on a semiconductor substrate may produce undesirable collapse or sticking issue. In the advanced technology nodes, the height of the transistor gates needs to be very higher. For example, when the gate length is less than 20 nm, the gate aspect ratio, defined as the gate height over the gate width, can be greater than 15. The high gate aspect ratio may cause adjacent gates to collapse or stick

together, especially during various processes, such as wet etching and cleaning. Other concerns includes under-cutting to shallow trench isolation (STI) during source/drain epitaxy growth or over-etching to the STI during forming contacts to source and drain in various examples.

**[0004]** Therefore, while existing fabrication techniques have been generally adequate for planar devices, in order to continue to meet ever-increasing design requirements, further advances are needed.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0005]** Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

**[0006]** FIG. 1A is a perspective view of a semiconductor structure, constructed in accordance with some embodiments.

**[0007]** FIG. 1B is a sectional view of the semiconductor structure of FIG. 1A along the dashed line AA', constructed in accordance with some embodiments.

**[0008]** FIG. 1C is a sectional view of the semiconductor structure of FIG. 1A along the dashed line BB', constructed in accordance with some embodiments.

**[0009]** FIG. 2 is a flowchart of an integrated circuit fabrication method in accordance with some embodiments.

**[0010]** FIG. 3 is a perspective view of a semiconductor structure, constructed in accordance with some embodiments.

**[0011]** FIGS. 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, 15A, and 16A are perspective views of the semiconductor structure at various fabrication stages, constructed in accordance with some embodiments.

**[0012]** FIGS. 4B, 5B, 6B, 7B, 8B, 9B, 10B, 11B, 12B, 13B, 15B, and 16B are top views of the semiconductor structure at various fabrication stages, constructed in accordance with some embodiments.

**[0013]** FIGS. 4C, 5C, 6C, 7C, 8C, 9C, 10C, 11C, 12C, 13C, 15C, and 16C are sectional views of the semiconductor structure at various fabrication stages, constructed in accordance with some embodiments.

**[0014]** FIG. 14 is a flowchart of an integrated circuit fabrication method in accordance with some embodiments.

## DETAILED DESCRIPTION

**[0015]** The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself

dictate a relationship between the various embodiments and/or configurations discussed.

**[0016]** Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as being “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary term “below” can encompass both an orientation of above and below. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

**[0017]** The present disclosure is directed to, but not otherwise limited to, a fin-like field-effect transistor (FinFET) device. The FinFET device, for example, may be a complementary metal-oxide-semiconductor (CMOS) device including a P-type metal-oxide-semiconductor (PMOS) FinFET device and an N-type metal-oxide-semiconductor (NMOS) FinFET device. The following disclosure will continue with a FinFET example to illustrate various embodiments of the present invention. It is understood, however, that the application should not be limited to a particular type of device, except as specifically claimed.

**[0018]** FIG. 1A is a perspective view of a portion of a workpiece 100 according to various aspects of the present disclosure. FIGS. 1B and 1C are sectional views of the workpiece 100 along the dashed lines AA' and BB', respectively, constructed according to various aspects of the present disclosure. FIGS. 1A–1C have been simplified for the sake of clarity and to better illustrate the concepts of the present disclosure. Additional features may be incorporated into the workpiece 100, and some of the features described below may be replaced or eliminated for other embodiments of the workpiece 100.

**[0019]** The workpiece 100 includes a substrate 102. The substrate 102 includes a bulk silicon substrate. Alternatively, the substrate 102 may include an elementary semiconductor, such as silicon or germanium in a crystalline structure; a compound semiconductor, such as silicon germanium, silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; or combinations thereof. Possible substrates 102 also include a silicon-on-insulator (SOI) substrate. SOI substrates are fabricated using separation by implantation of oxygen (SIMOX), wafer bonding, and/or other suitable methods. The substrate 102 also includes various isolation features, such as shallow trench isolation (STI) features 112 and active regions 106 defined by the STI features 112.

**[0020]** In some embodiments, the active regions are extended above the isolation features 112 with non-planar structure, such as a fin structure 106. The fin structure has one or more raised active regions (or fin active regions) separated by the STI features. Those fin active regions are also referred to as device fins 106, to be differentiated from the fill fins introduced later. The fin structure 106 is also referred to as device fin structure. While the illustrated

embodiments include a fin structure, further embodiments include other raised active and passive devices formed upon the substrate 102.

**[0021]** The workpiece 100 includes FETs, such as n-channel field effect transistor (nFET) and a p-channel field effect transistor (pFET). In some one example that the active regions are defined in the fin structure, the workpiece 100 includes FinFETs, such as exemplary an n-channel FinFET and a p-channel FinFET.

**[0022]** In the present embodiment, the workpiece 100 includes various FETs 152 designed for various functions, such as core devices, memory devices, and input/output (I/O) devices accordingly. Those devices and the formation will be described through the detailed description. One FET is illustrated by a dashed circle to include various features of the FET. In turn, each of FETs 152 includes a pair of opposing source/drain features (or source/drain regions) 136, which may include various doped semiconductor materials, a gate stack 146, and a channel region 154 (as a portion of the active region) disposed between the source/drain features and underlying the gate stack. The flow of carriers (electrons for the n-channel device and holes for the p-channel device) through the channel region 154 is controlled by a voltage applied to a gate stack 146 adjacent to the channel region (and overwrapping the channel region when the active regions are defined in a fin structure).

**[0023]** In some embodiments, the active regions are defined in the fin structure 106, the channel region 154 rises above the STI features 112. The raised channel region 154 provides a larger surface area proximate to the gate stack 146 than comparable planar devices. This strengthens the electromagnetic field interactions between the gate stack 146 and the channel region 154, which may reduce threshold voltage, leakage and short channel effects associated with smaller devices. Thus in many embodiments, FinFETs and other nonplanar devices deliver better performance in a smaller footprint than their planar counterparts.

**[0024]** In the present example, the device fins 106 have elongated shape oriented in the Y direction and the gate stacks 146 have elongated shape oriented in the X direction. The X and Y directions are orthogonal from each other. The workpiece 100 further includes fill fins 118, or collectively a fill fin structure. The fill fins 118 extruded above the isolation regions 112 but are dielectric features filling between the device fins 106. The fill fins 118 have various dimensions and orientations. For example, the fill fins 118 include fill fins 118A, 118B and 118C oriented in the Y direction with different widths. Those fill fins tune overall fin pattern density, reinforce the mechanical strength of the device fins, and enhance the manufacturing capability. The fill fins 118 also include fill fin 118D oriented in the X direction and serves as gate cut feature, so to improve the quality of the gate patterning through this double patterning technique. The fill fins 118 include dual dielectric material layers 120 and 122. In the present embodiments, the first fill dielectric material layer 120 is deposited by atomic layer deposition (ALD) and the second fill dielectric material layer 122 is deposited by spin-on coating or flowable chemical vapor deposition (FCVD). The first and second dielectric materials have different compositions. For examples, the first fill dielectric material layer 120 is a high-k dielectric material, such as hafnium oxide ( $\text{HfO}_2$ ), or zirconium oxide ( $\text{ZrO}_2$ ) while the second fill dielectric material layer 122 includes carbon-containing dielectric



material, such as silicon carbon oxide, silicon carbon oxynitride or silicon carbon nitride. For those fill fins with smaller widths, such as fill fins **118A** and **118B**, only the first dielectric material layer **120** presents. For those fill fins with greater widths, such as fill fins **118C** and **118BD**, both the first fill dielectric material layer **120** and the second fill dielectric material layer **122** present. By using the dual dielectric material layers to form the fill fins **118**, the processing capability is increased. Especially, after the first fill dielectric material layer **120**, the aspect ratios of the trenches are reduced to be easily filled by the second dielectric material layer **122**, thus the processing window is enhanced, which will be further explained later. The workpiece **100** may include other features and structures, such as contact features **150** to the source/drain features **136** and the metal gate cut features **148**. Those features and the method to form the same are further described through the following descriptions.

**[0025]** In some embodiments, the structure and the method of the present disclosure may provide the channel regions **154** that is un-doped and has higher carrier mobility and further enhanced device performance, such as increased device speed. In some other embodiments, the channel region **154** may be alternatively doped with proper types of dopants.

**[0026]** Exemplary methods of forming the workpiece **100** having FET devices will now be described with reference to FIGS. 2-16C. FIG. 2 is a flowchart of a method **200** for fabricating FET devices on a workpiece **100** according to various aspects of the present disclosure. The figures that follow refer perspective views of the workpiece **100**, top views, and/or cross-sections taken through the gate stacks **146** (e.g., along the dashed line AA' of FIG. 1A) or through the source/drain regions **136** (e.g., along the dashed line BB' of FIG. 1A) of the workpiece **100**.

**[0027]** The method **200** and the structure of the workpiece **100** are collectively described according to various aspects of the present disclosure. It is understood that additional steps can be provided before, during, and after the method **200** and that some of the steps described can be replaced or eliminated for other embodiments of the method. In the following description, the active regions are also collectively referred to as fin structure **106**. However, active regions are not limited to fin active regions in the fin structure.

**[0028]** Referring first to block **202** of FIG. 2 and to FIG. 3, a workpiece **100** is received, which includes a substrate **102**, such as a silicon substrate. The substrate **102** may alternatively or additionally include an elementary (single element) semiconductor, such as silicon or germanium in a crystalline structure; a compound semiconductor, such as silicon germanium, silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; a non-semiconductor material, such as soda-lime glass, fused silica, fused quartz, and/or calcium fluoride (CaF<sub>2</sub>); or combinations thereof.

**[0029]** The substrate **102** may be uniform in composition or may include various layers. The layers may have similar or different compositions, and in various embodiments, some substrate layers have non-uniform compositions to induce device strain and thereby tune device performance. Examples of layered substrates include silicon-on-insulator (SOI) substrates **102**. In some such examples, the substrate

**102** may include an embedded insulating layer such as a silicon oxide, a silicon nitride, a silicon oxynitride, or other suitable insulating materials.

**[0030]** The received workpiece **100** may have one or more layers formed upon it. For example, the substrate **102** includes one or more semiconductor layer epitaxially grown on bulk silicon, such as a silicon wafer. For example, the substrate **102** includes a first semiconductor layer formed on bulk silicon and a second semiconductor layer formed on the first semiconductor layer. The first semiconductor layer includes a first semiconductor material (such as SiGe) and the second semiconductor layer includes a second semiconductor material (such as Si) different from the first semiconductor material. The first and second semiconductor layers are epitaxially grown by suitable techniques, such as selective epitaxy growth (SEG). In some embodiments, suitable deposition processes for epitaxy growth include ALD, chemical vapor deposition (CVD), high-density plasma CVD (HDP-CVD), physical vapor deposition (PVD) and/or other suitable deposition processes. Any of these techniques may be used to grow the semiconductor layer having any composition including a graded composition.

**[0031]** Various doping process may be applied to the substrate **102** through a proper method, such as ion implantation. In the present embodiment, an anti-punch-through (APT) process is applied to the substrate **102** to introduce proper dopants to respective regions of the substrate through ion implantations. The APT process may include forming a hard mask with openings defining regions for nFETs; performing an ion implantation to the nFET regions; and removing the hard mask, and similar steps for pFETs.

**[0032]** To facilitate fabrication and to avoid damage to the semiconductor layer, one or more hard mask layer **104** may be formed on the substrate **102**. For example, the hard mask layer **104** includes a dielectric such as semiconductor oxide, semiconductor nitride, semiconductor oxynitride, or semiconductor carbide. In some examples, the hard mask layer **104** includes two or more films stacked together, such as a silicon oxide film and a silicon nitride film in stack. The hard mask layer **104** may be formed by thermal growth, ALD, CVD, HDP-CVD, PVD, and/or other suitable deposition processes. The hard mask may include other suitable material, such as a silicon oxide layer and a poly-silicon layer on the silicon oxide layer.

**[0033]** Referring to an operation **204** of FIG. 2 and FIGS. 4A-4C, the method **200** patterns the substrate **102** to form one or more device fins **106** extending from the substrate **102**. FIG. 4A is a schematic view of the workpiece **100**; FIG. 4B is a top view of the workpiece **100**; and FIG. 4C is a sectional view of the workpiece **100** along the dashed line AA'. In some embodiments, the operation **204** includes lithography process and etching. In furtherance of the embodiments, the operation **204** includes forming a patterned photoresist (or resist) layer by a lithography process and etching to form trenches and a fin structure using the patterned resist layer as an etch mask. In the present embodiment, the openings in the patterned resist layer are first transferred to the hard mask **104** by a first etching and then are transferred to the substrate **102** by a second etching. More details of the operation **204** are further provided below.

**[0034]** A resist used to define the fin structure **106** may be formed on the hard mask layer **104**. An exemplary resist layer includes a photosensitive material that causes the layer

to undergo a property change when exposed to light, such as ultraviolet (UV) light, deep UV (DUV) light or extreme UV (EUV) light. This property change can be used to selectively remove exposed or unexposed portions of the resist layer by a developing process referred. This procedure to form a patterned resist layer is also referred to as lithographic patterning or lithography process.

**[0035]** In one embodiment, the resist layer is patterned to leave the portions of the photoresist material disposed over the workpiece **100** by the lithography process. After patterning the resist, an etching process is performed on the workpiece **100** to open the hard mask layer **104**, thereby transferring the pattern from the resist layer to the hard mask layer **104**. The remaining resist layer may be removed after the patterning the hard mask layer **104**. An exemplary lithography process includes spin-on coating a resist layer, soft baking of the resist layer, mask aligning, exposing, post-exposure baking, developing the resist layer, rinsing, and drying (e.g., hard baking). Alternatively, a lithographic process may be implemented, supplemented, or replaced by other methods such as maskless photolithography, electron-beam writing, and ion-beam writing. The etching process to pattern the hard mask layer may include wet etching, dry etching or a combination thereof. The first etching process applied to the hard mask **104** may include multiple etching steps. For example, the silicon oxide film in the hard mask layer may be etched by a diluted hydrofluorine solution and the silicon nitride film in the hard mask layer may be etched by a phosphoric acid solution. The second etching process applied to the substrate **102** may include any suitable etching technique such as dry etching, wet etching, other etching methods (e.g., reactive ion etching (RIE)), or a combination thereof. In some examples, the second etching process may include multiple etching steps with different etching chemistries, each targeting a particular material of the workpiece **100**. In some examples, the semiconductor material of the substrate may be etched by a dry etching process using a fluorine-based etchant. In some embodiments, etching includes multiple etching steps with different etching chemistries, each targeting a particular material of the substrate **102** and each selected to resist etching the hard mask **104**. For example, a dry etching process may implement an oxygen-containing gas, a fluorine-containing gas (e.g.,  $\text{CF}_4$ ,  $\text{SF}_6$ ,  $\text{CH}_2\text{F}_2$ ,  $\text{CHF}_3$ , and/or  $\text{C}_2\text{F}_6$ ), a chlorine-containing gas (e.g.,  $\text{Cl}_2$ ,  $\text{CHCl}_3$ ,  $\text{CCl}_4$ , and/or  $\text{BCl}_3$ ), a bromine-containing gas (e.g.,  $\text{HBr}$  and/or  $\text{CHBR}_3$ ), an iodine-containing gas, other suitable gases and/or plasmas, and/or combinations thereof. For example, a wet etching process may comprise etching in diluted hydrofluoric acid (DHF); potassium hydroxide (KOH) solution; ammonia; a solution containing hydrofluoric acid (HF), nitric acid ( $\text{HNO}_3$ ), and/or acetic acid ( $\text{CH}_3\text{COOH}$ ); or other suitable wet etchant. The remaining portions of the semiconductor layers become the device fins **106**, defining the trenches **108** between the device fins **106**, such as the trenches **108A**~**108D** in the illustrated embodiment.

**[0036]** The etching processes are designed to produce fin structure **106** of any suitable height and width extending from the substrate **102**. Particularly, the etching process applied to the substrate **102** is controlled such that the substrate **102** is partially etched, as illustrated in FIG. 4A. This may be achieved by controlling etching time or by controlling other etching parameter(s). By the etching processes, the fin structure is formed and extended from the

substrate **102**. The fin structure **106** includes a plurality of elongated fin-like active regions (simply fins) oriented in the Y direction. In addition to defining the fin structure **106**, the etching process also define one or more isolation trench **108** between the active regions of the fin structure **106**. The fin active regions of the fin structure **106** are also referred to as device fins, to differentiate from fill fins to be introduced later. As illustrated from the above description, the device fins **106** include one or more semiconductor material same or different form that of the substrate **102**. For example, the device fins **106** include silicon, germanium, silicon germanium, or other suitable semiconductor material. In some other examples, the device fins **106** include silicon germanium with graded concentration, such as the germanium concentration increasing toward the top surface of the device fins.

**[0037]** Referring to FIG. 2 and FIGS. 5A-5C, the method **200** proceeds to an operation **206** to form a first dielectric layer **112** on the substrate **102**. FIG. 5A is a schematic view of the workpiece **100**; FIG. 5B is a top view of the workpiece **100**; and FIG. 5C is a sectional view of the workpiece **100** along the dashed line AA'. In the present embodiment, the dielectric layer **112** is deposited on the substrate **102** with a conformal profile, covering the fin structure **106**. The dielectric layer **112** may include a single dielectric material layer or multiple dielectric material layers. Suitable dielectric materials for the dielectric layer **112** include silicon oxides, silicon nitrides, silicon carbides, fluoro-silicate glass (FSG), low-K dielectric materials, other suitable dielectric materials, or a combination thereof. The dielectric material may be deposited by any suitable technique including thermal growth, CVD, HDP-CVD, PVD, ALD, and/or spin-on techniques. In the illustrated embodiment, an ALD process is used as conformal deposition technique.

**[0038]** The dielectric layer **112** constitutes, at least partially, an isolation feature, such as a shallow trench isolation feature (STI). The isolation feature may include multiple layers, such as the dielectric layer **112** depicted in FIG. 5A as one of the multiple layers.

**[0039]** As will be shown below at later stage, fill fins will be formed in some of these trenches. In contrast, some trenches between device fins **106** with relatively narrow spacing (e.g.,  $<2$  nm) may be filled up by the dielectric layer **112**, such as the trench **108A**, and thus no fill fin is formed in this trench. Some trenches may have a reduced width substantially same to a width of the device fin **106**, such as the trench **108B**. Some trenches between device fins **106** with relatively wide spacing may still have large gap, such as the trench **108C** or **108D**.

**[0040]** Referring to FIG. 2 and FIGS. 6A-6C, the method **200** proceeds to an operation **208** to form a course cut (or cut trench) **114** to the devices fins **106** by lithography patterning and etching. FIG. 6A is a schematic view of the workpiece **100**; FIG. 6B is a top view of the workpiece **100**; and FIG. 6C is a sectional view of the workpiece **100** along the dashed line AA'. The cut trench **114** is a trench cut through the device fins **106**, defining ends of the device fins **106**. The cut trench **114** extends through multiple device fins **106** along the X direction. The lithography patterning and etching in the operation **208** is similar to those implemented in the operation **204**. Particularly, the patterned resist layer **116** is formed by a lithography patterning process, and an etching process is applied to the device fins **106** and the dielectric layer **112** to form the cut trench **114**. The cut trench **114** has

a width  $W_c$  substantially greater than the width  $W_f$  of the device fins **106**. In some embodiments, the ratio  $W_c/W_f$  is greater than 2, such as ranging from 3 to 4. After the formation of the cut trench **114**, the resist layer **116** is removed by wet stripping or plasma ashing.

**[0041]** Referring to FIG. 2 and FIGS. 7A-7C, the method **200** proceeds to an operation **210** to fill the trenches of the device fins, thereby forming fill fins **118** (such as fill fins **118A**~**118C**) on the substrate **102**. FIG. 7A is a schematic view of the workpiece **100**; FIG. 7B is a top view of the workpiece **100**; and FIG. 7C is a sectional view of the workpiece **100** along the dashed line AA'. The fill fins **118** are dielectric features extended vertically on the substrate **102**. The fill fins **118** do not function as active regions but for other purposes, such as tuning patterning density for better fabrication, therefore are referred differently from the device fins. The fill fins **118** are deposited in the trenches **108**, such as the fill fin **118A** in the trench **108B**; the fill fin **118B** in the trench **108C**; and the fill fin **118C** in the trench **108D**. The fill fins **118** include multiple dielectric material layers. In the present embodiment, the fill fins **118** include a first fill dielectric layer **120** and a second fill dielectric layer **122**. In the furtherance of the embodiment, the first fill dielectric layer **120** is deposited by ALD, and the second fill dielectric layer **122** is deposited by flowable CVD (FCVD) or alternatively by spin-on coating. As noted above, the trenches **108** have varying width. Some trenches (such as trenches **108B** and **108C**) have smaller dimensions (e.g., less than 40 nm) and are completely filled by the first fill dielectric layer **120**. Some trenches (such as trench **108D**) have greater dimensions (e.g., greater than 40 nm) and are completely filled by both the first fill dielectric layer **120** and the second fill dielectric layer **122**. For the trenches with the greater dimensions, the first fill dielectric layer is conformal to the trenches. Thus, the trenches with larger dimensions have dual fill dielectric layers while the trenches with smaller dimensions have a single fill dielectric layer. In some examples, the first fill dielectric layer **120** has a thickness ranging between 1 nm and 20 nm. In some examples, the first fill dielectric layer **120** has a thickness ranging between 1 nm and 3 nm.

**[0042]** The fill dielectric layers may chose suitable dielectric materials different from that of the dielectric material layer **112** to achieve etching selectivity at later etching stage. The first and second dielectric materials have different compositions. For examples, the first fill dielectric material layer **120** is a high-k dielectric material, such as hafnium oxide ( $HfO_2$ ), or zirconium oxide ( $ZrO_2$ ) while the second dielectric material layer **122** includes carbon-containing dielectric material, such as silicon carbon oxide, silicon carbon oxynitride or silicon carbon nitride. In other embodiments, the first fill dielectric layer **120** may include other suitable dielectric material, such as metal oxide (such as aluminum oxide  $Al_2O_3$ ), or metal nitride (such as aluminum nitride  $AlN$ ), or a combination thereof. The second fill dielectric layer **122** may include other dielectric layer, such as silicon oxide, silicon oxynitride, silicon carbide nitride, and/or other suitable dielectric materials. In one example, the first fill dielectric layer **120** includes hafnium oxide deposited by ALD, and the second fill dielectric layer **122** includes silicon carbon oxide deposited by FCVD or spin-on deposition. In some examples, the fill fins of smaller widths (such as **118A** with width  $W_d$ ) have substantially the same width as the width  $W_f$  of the device fins **106**. As described

above, fill fins **118** have different dimensions. For examples, the fill fins **118B** and **118C** have dimensions greater than that of the fill fin **118A**.

**[0043]** In the operation **210**, the cut trench **114** is filled as well by both the fill dielectric layers **120** and **122**, thereby forming another fill fin **118D** (also referred to as fin cut feature) in the cut trench **114**. The fill fin **118D** is extended and oriented in the X direction, which is orthogonal to the direction of the other fill fins (such as **118A**, **118B** and **118C**) and the device fins. Particularly, the fin cut feature **118D** has a width different from the widths of the other fill fins **118A**~**118C** since it is defined by a separate patterning process in the operation **208** and is designed for different criteria (such as enough spacing between adjacent fin ends).

**[0044]** Referring to FIG. 2 and FIGS. 8A-8C, the method **200** proceeds to an operation **212** to perform a polishing such as chemical mechanic polishing (CMP) process to the fill fins **118** to planarize the top surface and remove the excessive portions of the fill dielectric layers **120** and **122** deposited on device fins **106**. FIG. 8A is a schematic view of the workpiece **100**; FIG. 8B is a top view of the workpiece **100**; and FIG. 8C is a sectional view of the workpiece **100** along the dashed line AA'. In some embodiments, the hard mask **102** may function as a CMP stop layer. In some embodiments, the hard mask **102** may be removed by the CMP process or an additional etching process.

**[0045]** When applying only one deposition technique, such as FCVD or spin-on coating, to fill the trenches to form the fill fins, the bottom portions of the fill fins have poor quality, especially for the fill fins with smaller dimensions or greater aspect ratios, which limits the processing window to a smaller range of aspect ratios. In the disclosed method, both the first fill dielectric layer **120** by ALD and the second dielectric layer **122** by FCVD are implemented to fill the trenches and to form the fill fins. As described above, the trenches **108B**, **108C** and **108D** are all completely filled by the disclosed method. By implementing the bi-layer fill fins and the corresponding method to fill the trenches among the device fins, it is achieved to fill all trenches with relative aspect ratio reduced to a smaller range, therefore increasing the processing windows, the circuit performance and the manufacturing capability. Here the aspect ratio of a trench is defined as the depth (D) over width (W) of the trench, as formulated as  $AS=D/W$ . The follow descriptions will take the above three fill fins **118A**, **118B** and **118C** (corresponding to trenches **108B**, **108C** and **108D**) as an example. In the present example, the trenches **108B** and **108C** are completely filled by the first fill dielectric layer **120** while the trench **108D** is collectively filled by the dual fill dielectric layers (**120** and **122**). In the present embodiment, the first fill dielectric layer **120** has a thickness  $T_1$  ranging between 12 nm and 25 nm; and the second fill dielectric layer **122** in the trench **108D** has a thickness  $T_2$  ranging between 16 nm and 360 nm, as illustrated in FIG. 8C. Furthermore, the thickness ratio  $T_2/T_1$  of the dual fill dielectric layers in the trench **118D** is greater than 1.1. In some examples, the thickness ratio  $T_2/T_1$  ranges between 1.2 and 15.

**[0046]** If only the second dielectric layer **122** (by FCVD or spin-on coating) is used to fill those trenches and to form the corresponding fill fins, the method is capable of completely filling the trench **108B**, or other trenches with aspect ratios less than that of the trench **108B**. By using the dual fill dielectric layers to form the fill fins, the trenches **108** to be collectively filled by the dual fill dielectric layers after

deposition of the first fill dielectric layer **120** have aspect ratios reduced to a smaller range. Accordingly, when the second fill dielectric layer **122** is filled to those trenches, such as **108D**, the aspect ratios of the corresponding trenches are reduced to less ones, and the second fill dielectric layer **122** is easily filled in with high quality. In one example for illustration, the aspect ratio of the trenches for the second fill dielectric layer **122** is reduced from about 15 to about 9.3. Therefore, when the disclosed bi-layer fill fins and the corresponding method are implemented, the aspect ratio requirement is substantially relaxed; the processing window is enlarged; and the manufacturing capability is enhanced.

[0047] Referring to FIG. 2 and FIGS. 9A-9C, the method **200** proceeds to an operation **214** by selectively recessing the dielectric layer **112**. FIG. 9A is a schematic view of the workpiece **100**; FIG. 9B is a top view of the workpiece **100**; and FIG. 9C is a sectional view of the workpiece **100** along the dashed line AA'. After the dielectric layer **112** is recessed, the fin structure **106** and fill fins **118** are extruded above the recessed dielectric layer **112**, with trenches **124** defined between the device fins **106** and the fill fins **118**. The device fins **106** are electrically isolated from each other by the recessed dielectric layer **112**, which functions as isolation features. Any suitable etching technique may be used to recess the dielectric layer **112**, including dry etching, wet etching, RIE, and/or other etching methods. In an exemplary embodiment, an anisotropic dry etching is used to selectively remove the dielectric layer **112** without etching the fin structure **106**, with a proper etching gas, such as fluorine-containing or chlorine-containing gas. The height of the device fins **106** are determined by the etching depth of the etching process used to recess the dielectric layer **112**.

[0048] Referring to FIG. 2 and FIGS. 10A-10C, the method **200** may proceed to an operation **216** by forming a dielectric layer **126** on the device fins **106** and the fill fins **118**. FIG. 10A is a schematic view of the workpiece **100**; FIG. 10B is a top view of the workpiece **100**; and FIG. 10C is a sectional view of the workpiece **100** along the dashed line AA'. The dielectric material layer **216** may be formed on the fin structure **106** within the I/O device region and serves as the gate dielectric layer for the I/O FinFETs, or as a part of the gate dielectric layer of the I/O FinFETs so that the I/O FinFETs have a thicker gate dielectric layer for robust performance due to more harsh power surge on the I/O devices. In some embodiments, the dielectric layer **216** includes silicon oxide deposited by a suitable method, such as ALD, CVD, thermal oxidation, ozone oxidation, etc. The formation of the dielectric layer **126** may further include a subsequent annealing process to improve the material quality, such as increasing the material density and reducing the defects. In the present embodiment, the dielectric material layer is deposited on the fin structure in both core device region and I/O device region, and is subsequently removed from the core device region after dummy gates are removed. Furthermore, the dielectric layer **126** is a conformal layer to the profile of the device fins **106** and the fill fins **118**. In the top view of the workpiece **100**, the dielectric layer **126** covers the device fins **106** and the fill fins **118**. For illustrative purpose, those device fins **106** and fill fins **118** are outlined in FIG. 10B, with dashed lines and solid lines, respectively.

[0049] Referring to FIG. 2 and FIGS. 11A-11C, the method **200** proceeds to an operation **218** by forming gate stacks **130**, such as exemplary gate stacks **130A-130E**. FIG.

11A is a schematic view of the workpiece **100**; FIG. 11B is a top view of the workpiece **100**; and FIG. 11C is a sectional view of the workpiece **100** along the dashed line AA'. In the present embodiment, the gate stacks **130** will be replaced by metal gate stacks at later fabrication stage, therefore also be referred to as dummy gate stacks. The dummy gate stacks **130** are formed over the channel regions of the device fins **106**. In some examples, the formation of the dummy gate stacks **130** includes depositing a dummy gate layer containing polysilicon or other suitable material and patterning the layer. A gate hard mask layer **132** may be formed on the dummy gate material layer and is used as an etch mask during the patterning the dummy gate layer. The gate hard mask layer **132** may include any suitable material, such as a silicon oxide, a silicon nitride, a silicon carbide, a silicon oxynitride, other suitable materials, and/or combinations thereof. In one embodiment, the gate hard mask **132** includes dual mask material films **132A** and **132B**, such as silicon oxide and silicon nitride. In some embodiments, the patterning process to form the dummy gate stacks includes forming a patterned resist layer by lithography process; etching the hard mask layer using the patterned resist layer as an etch mask; and etching the dummy gate layer to form the dummy gate stacks using the patterned hard mask layer as an etch mask.

[0050] In some embodiments, one or more gate sidewall feature (gate spacer) **134** is formed on the sidewalls of the dummy gate stacks **130**, as illustrated in FIGS. 12A-12C. The gate sidewall feature **134** may be used to offset the subsequently formed source/drain features and may be used for designing or modifying the source/drain profile. The gate sidewall feature **134** may include any suitable dielectric material, such as a semiconductor oxide, a semiconductor nitride, a semiconductor carbide, a semiconductor oxynitride, other suitable dielectric materials, and/or combinations thereof. In some embodiments, the gate sidewall feature **134** may include multiple layers, such as a first gate spacer (or a seal layer) on sidewalls of the dummy gate stacks **130** and a second gate spacer on the first gate spacer. In furtherance of the embodiments, the first gate spacer is silicon oxide and the second gate spacer is silicon nitride. In one example, the gate sidewall features are formed by deposition and anisotropic etching, such as dry etching. In another example, the first gate spacer is formed by ALD and the second gate spacer is formed by deposition and anisotropic etching.

[0051] Referring to FIG. 2 and FIGS. 12A-12C, the method **200** proceeds to an operation **220** by forming epitaxial source/drain features **136** within the source/drain regions, which are defined in the fin structure and on both sides of the gate stacks **130**. FIG. 12A is a schematic view of the workpiece **100**; FIG. 12B is a top view of the workpiece **100**; and FIG. 12C is a sectional view of the workpiece **100** along the dashed line AA'. The epitaxial source/drain features **136** may be formed by selective epitaxy growth for strain effect with enhanced carrier mobility and device performance. The dummy gate **130** and gate spacer **134** limit and constrain such that the source/drain features **136** are self-aligned in the source/drain regions. In many embodiments, the source/drain features **136** are formed by one or more epitaxial growth (epitaxial process), whereby silicon (Si) features, silicon germanium (SiGe) features, silicon carbide (SiC) features, and/or other suitable semiconductor features are grown in a crystalline state on

the fin structure **106** within the source/drain regions. In an alternative embodiment, an etching process is applied to recess portions of the fin structure **106** within the source/drain regions before the epitaxy growth. The etching process may also remove any dielectric material disposed on the source/drain regions, such as during the formation of the gate sidewall features. Suitable epitaxy processes include CVD deposition techniques (e.g., vapor-phase epitaxy (VPE) and/or ultra-high vacuum CVD (UHV-CVD), molecular beam epitaxy, and/or other suitable processes.

**[0052]** The source/drain features **136** may be in-situ doped during the epitaxy process by introducing doping species including: p-type dopants, such as boron or  $\text{BF}_2$ ; and n-type dopants, such as phosphorus or arsenic. If the source/drain features **110** are not in-situ doped, an implantation process (i.e., a junction implant process) is performed to introduce the corresponding dopant into the source/drain features **136**. In an exemplary embodiment, the source/drain features **136** in an nFET include silicon doped with phosphorous (SiP) or silicon carbide doped with phosphorous (SiCP), while those in a pFET include silicon germanium doped with boron (SiGeB), SiGeSnB (tin may be used to tune the lattice constant) and/or GeSnB. In some other embodiments, the raised source/drain features **136** include more than one semiconductor material layers. For example, a silicon germanium layer is epitaxially grown on the substrate within the source/drain regions and a silicon layer is epitaxially grown on the silicon germanium layer. One or more annealing processes may be performed thereafter to activate the source/drain features **110**. Suitable annealing processes include rapid thermal annealing (RTA), laser annealing processes, other suitable annealing technique or a combination thereof.

**[0053]** Referring to FIG. 2 and FIGS. 13A-13C, the method **200** proceeds to an operation **222**, in which an inter-level dielectric material (ILD) **140** is formed on the substrate to cover the source/drain features **136** in the source/drain regions. FIG. 13A is a schematic view of the workpiece **100**; FIG. 13B is a top view of the workpiece **100**; and FIG. 13C is a sectional view of the workpiece **100** along the dashed line AA'. The ILD **140** acts as an insulator that supports and isolates conductive traces (such as contacts, vias and metal lines) to be formed therein. The ILD **140** may include any suitable dielectric material, such as a silicon oxide, low-k dielectric material, porous dielectric material, other suitable dielectric material or a combination thereof. In an alternative embodiment, an etch stop layer **142** is deposited on the substrate before the forming of the ILD **140**. The etch stop layer **142** functions as etch stopping during the etching to form contacts in the ILD during later fabrication stages. The etch stop layer **142** includes a material different from that of the ILD **140** to provide etch selectivity. For example, the etch stop layer **142** may include silicon nitride deposited by CVD or ALD. In some embodiments, the formation of the ILD **1102** includes deposition and CMP to provide a planarized top surface. The hard mask **132** may be removed during the CMP process, an additional etching operation, or a combination thereof.

**[0054]** Still referring to FIG. 2 and to FIGS. 13A-13C, the method **200** proceeds to an operation **224** to form metal gate stacks **146** (such as metal gate stacks **146A-146E**) to replace the dummy gate stacks **130**. In the operation **224**, the dummy gate stacks **130** are removed by suitable selective etching (such as wet etching), resulting in gate trenches. The etching

process may include multiple etching steps to remove the dummy gate if more materials present. After the removal of the dummy gate stacks **130**, metal gate materials are deposited in the gate trenches, and a CMP process is applied to remove the excessive gate materials and planarize the top surface.

**[0055]** The gate materials of the metal gate stacks **146** include gate dielectric layer and gate electrode. In some embodiments, the gate dielectric layer includes a high-k dielectric material, and the gate electrode includes metal or metal alloy. The metal gate stacks **146** are formed on the workpiece **100** wrapping around the channel regions of the fin structure **106**. In some examples, the gate dielectric layer and the gate electrode each may include a number of sub-layers. The high-k dielectric layer may include metal oxide, metal nitride, such as  $\text{LaO}$ ,  $\text{AlO}$ ,  $\text{ZrO}$ ,  $\text{TiO}$ ,  $\text{Ta}_2\text{O}_5$ ,  $\text{Y}_2\text{O}_3$ ,  $\text{SrTiO}_3$  (STO),  $\text{BaTiO}_3$  (BTO),  $\text{BaZrO}$ ,  $\text{HfZrO}$ ,  $\text{HfLaO}$ ,  $\text{HfSiO}$ ,  $\text{LaSiO}$ ,  $\text{AlSiO}$ ,  $\text{HfTaO}$ ,  $\text{HfTiO}$ ,  $(\text{Ba},\text{Sr})\text{TiO}_3$  (BST),  $\text{Al}_2\text{O}_3$ ,  $\text{Si}_3\text{N}_4$ , oxynitrides (SiON), or other suitable dielectric materials. The high-k dielectric layer is deposited a suitable technique, such as ALD, CVD, metal-organic CVD (MOCVD), PVD, thermal oxidation, and/or other suitable techniques. The gate dielectric layer may additionally include an interfacial layer disposed between the fin and the high-k dielectric layer. The interfacial layer may include may include silicon oxide, silicon nitride, silicon oxynitride, and/or other suitable material, deposited by a suitable method, such as ALD, CVD, ozone oxidation, etc.

**[0056]** The gate electrode material is then filled in the gate trench. The gate electrode **1604** is formed by ALD, PVD, CVD, plating, other suitable process, or a combination thereof. The gate electrode may include a single layer or multiple layers, such as a metal layer, a liner layer, a wetting layer, and/or an adhesion layer. The gate electrode may include Ti, Ag, Al, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, TiN, TaN, Ru, Mo, Al, WN, Cu, W, or any suitable materials. In some embodiments, different metal materials are used for nFET and pFET devices with respective work functions, such as with a work function of 4.2 eV or less for nFET and a work function of 5.2 eV or greater for pFET. In some embodiments, the n-type WF metal includes tantalum (Ta). In other embodiments, the n-type WF metal includes titanium aluminum (TiAl), titanium aluminum nitride (TiAlN), or combinations thereof. In other embodiments, the n-metal include Ta, TiAl, TiAlN, tungsten nitride (WN), or combinations thereof. The n-type WF metal may include various metal-based films as a stack for optimized device performance and processing compatibility. In some embodiments, the p-type WF metal includes titanium nitride (TiN) or tantalum nitride (TaN). In other embodiments, the p-metal include TiN, TaN, tungsten nitride (WN), titanium aluminum (TiAl), or combinations thereof. The p-type WF metal may include various metal-based films as a stack for optimized device performance and processing compatibility. The work function metal is deposited by a suitable technique, such as PVD. In other examples, a barrier layer may be formed on deposited for lining the gate trenches before filling the metal. The barrier layer may include titanium, titanium nitride, tantalum, tantalum nitride or combination, deposited by suitable technique, such as PVD. In some examples, the gate dielectric layer includes the interfacial layer and the high-k dielectric layer. The gate electrode

includes a capping layer, a metal layer to tune the work function, and a filling metal, such as aluminum, copper or tungsten.

[0057] The method 200 may continue to include other operations described in FIG. 14. Referring to FIG. 14 and to FIGS. 15A-15C, the method 200 proceeds to an operation 226 to cut metal gate stacks 146. When the semiconductor technologies advance to higher technology node with smaller features and higher packing densities, the lithography process experiences challenges in its capability and resolution, the patterning of the gate stacks may be achieved by multiple patterning, such as double patterning. For example, the pattern of the gate stacks is decomposed into two patterning procedures, or two sets of lithography process and etching: one forming long gate stacks and another cutting the long gate stacks to shorter according to the design layout. In the present example, the metal gate stacks 146 are patterned and filled with one or more dielectric material to form gate cut features 148. The gate cut features 148 are dielectric features to define the gate ends and the spacing between the gate ends. The formation of the gate cut features 148 may include lithography process, etching and deposition, and may be further followed by CMP. For examples, the formation of the gate cut features includes a lithography process to form a patterned resist layer with openings that define the regions for gate cut features, performing an etching process to selectively etch the gate stacks through the resist openings to form trenches in the gate stacks, depositing one or more dielectric material to fill in the trenches; and performing a CMP process to remove the excessive the dielectric material. The dielectric material of the gate cut features may include silicon oxide, silicon nitride, low-k dielectric material, other suitable dielectric material, or a combination thereof.

[0058] Referring to FIG. 14 and to FIGS. 16A-16C, the method 200 may further include an operation 228 to form contact features 150. The contact features 150 are conductive features to electrically connect the FETs to form a functional circuit. In the present examples, the contact features 150 are designed to land on source/drain features 136. The formation of the contact features 150 may include lithography process, etching and deposition, and may be further followed by CMP. For examples, the formation of the contact features includes a lithography process to form a patterned resist layer with openings that define the regions for contact features, performing an etching process to the ILD 140 through the resist openings to form contact holes in the ILD 140, depositing one or more conductive material to fill in the contact holes; and performing a CMP process to remove the excessive the conductive material. The conductive material of the contact features may include tungsten, copper, aluminum, silicide, other metal or metal alloys, or a combination thereof. The contact features 150 may further include a barrier layer (such as titanium, titanium nitride, tantalum, tantalum nitride or a combination thereof) for lining the contact holes. For example, the contact features 150 include titanium and titanium nitride as a barrier layer deposited by PVD, and tungsten to fill in the contact holes by PVD, plating, other suitable technique or a combination thereof.

[0059] The method 200 may further include other operations during, before and/or after the described operations. For example, after the operation 228, the method 200 includes an operation 230 to form other features including

other interconnection features. An interconnection structure includes various conductive features to electrically connect various devices (including FETs) to an integrated circuit. The interconnection structure includes contact features, via features and metal lines. The metal lines may be distributed in multiple metal layers and via features vertically connect the metal lines between adjacent metal layers. For example, via features and metal lines may utilize copper technologies and may be formed a damascene process, such as dual damascene process or single damascene process.

[0060] The present disclosure provides a method making a semiconductor structure having various FinFETs and the semiconductor structure wherein the FinFETs have bi-layer fill fins. Depending on dimensions, some fill fins with smaller widths only include the first dielectric layer while some other fill fins with greater widths include both the first and second dielectric layers. Particularly, the first dielectric layer is deposited by ALD and the second dielectric layer is deposited using a flow mechanism, such as spin-on coating or FCVD. The fill fins may include some in an orientation parallel with the devices and some in an orthogonal orientation parallel with the gate electrodes. Various advantages may present in different embodiments. Some of the fill fins vary the pattern density of the device fins, therefore enhancing the mechanical strength of the fin structure without collapsing. Some of the fill fins may also serve to cut the gate stacks. Furthermore, the bi-layer fill fins and the corresponding method further provide an approach to fill the trenches (among the device fins) to a reduced range of the aspect ratio, therefore increasing the processing windows, the circuit performance and the manufacturing capability.

[0061] Thus, the present disclosure provides a semiconductor structure in accordance with some embodiments. The semiconductor structure includes device fins formed on a substrate; fill fins formed on the substrate and disposed among the device fins; and gate stacks formed on the device fins and the fill fins. The fill fins include a first dielectric material layer and a second dielectric material layer deposited on the first dielectric material layer. The first and second dielectric material layers are different from each other in composition.

[0062] The present disclosure provides a method for fabricating an integrated circuit. The method includes forming a device fin structure on a substrate, wherein the device fin structure includes a plurality of device fins and defines trenches among the device fins; filling the trenches with a first dielectric material layer and a second dielectric material layer, resulting in a fill fin structure having a plurality of fill fins; and forming gate stacks on the device fins and the fill fins.

[0063] The present disclosure provides a semiconductor structure. The semiconductor structure includes a plurality of device fins formed on a substrate; a plurality of fill fins formed in trenches among the device fins. The fill fins includes a first fill fin having a first width and a second fill fin with a second width greater than the first width. The first fill fin includes a first dielectric material layer. The second fill fin includes the first dielectric material layer and a second dielectric material layer. The semiconductor structure further includes gate stacks formed on the device fins and the fill fins. The first dielectric material layer includes at least one of metal oxide and metal nitride. The second dielectric material layer includes a dielectric layer containing silicon and carbon.

[0064] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor structure, comprising:
  - a first device fin and a second device fin formed on a substrate;
  - dielectric fins formed on the substrate, wherein the dielectric fins include a first dielectric fin spaced away from the first and second device fins, and a dielectric cut fin cutting through the first and second device fins; and
  - a gate stack formed on the first and second device fins and the first dielectric fin, wherein the first and second device fins, the first dielectric fin and the dielectric cut fin have a coplanar top surface.
2. The semiconductor structure of claim 1, further comprising a second dielectric fin, wherein
  - the first and second device fins and the first and second dielectric fins are longitudinally oriented along a first direction and are configured alternatively along a second direction being orthogonal to the first direction; and
  - the dielectric cut fin is longitudinally oriented along the second direction.
3. The semiconductor structure of claim 2, wherein
  - the first and second device fins and the first and second dielectric fins are spaced away from each other along the second direction, defining gaps each spanning between one of the first and second device fins and one of the first and second dielectric fins; and
  - portions of the gate stack are inserted in the gaps.
4. The semiconductor structure of claim 2, wherein
  - the first dielectric fin spans a first width along the second direction;
  - the second dielectric fin spans a second width along the second direction; and
  - the second width is greater than the first width.
5. The semiconductor structure of claim 4, wherein
  - the dielectric cut fin has a third width different greater than the second width; and
  - the dielectric cut fin directly contacts sidewalls of the first and second device fins.
6. The semiconductor structure of claim 2, wherein
  - each of the second dielectric fin and the dielectric cut fin includes both a first and second dielectric material layers;
  - the first dielectric fin includes the first dielectric material layer and is free of the second dielectric material layer; and
  - the first and second dielectric material layers are different from each other in composition.
7. The semiconductor structure of claim 6, wherein
  - the gate stack is longitudinally oriented in the second direction; and

the first dielectric material layer surrounds the second dielectric material layer and contacts sidewalls and bottom surfaces of the second dielectric material layer.

8. The semiconductor structure of claim 6, wherein
  - the first dielectric material layer includes a high-k dielectric material; and
  - the second dielectric material layer includes a carbon-containing material.
9. The semiconductor structure of claim 8, wherein
  - the first dielectric material layer is chosen from a metal oxide, a metal nitride and a combination thereof; and
  - the second dielectric material layer includes carbon and silicon.
10. The semiconductor structure of claim 6, wherein
  - the dielectric fins further include a third dielectric fin having a width that is greater than the first width and less than the second width;
  - the third dielectric fin includes the first dielectric material layer and is free of the second dielectric material layer; and
  - the third dielectric fin is longitudinally oriented along the first direction.
11. A method for fabricating an integrated circuit, comprising:
  - forming device fins longitudinally oriented along a first direction on a substrate with a first trench and a second trench defined among the device fins, wherein the first and second trenches spanning different widths along a second direction being orthogonal to the first direction;
  - forming a first dielectric material layer on the device fins and in the first and second trenches;
  - forming a cut trench on the device fins;
  - forming a second dielectric material layer in the first and second trenches, and the cut trench, thereby filling the first trench; and
  - forming a third dielectric material layer to fill in the second trench and the cut trench.
12. The method of claim 11, further comprising:
  - performing a chemical mechanical polishing process to remove the first, second and third dielectric material layers; and
  - selectively removing upper portions of the first dielectric material layer, resulting in a first dielectric fin and a second dielectric fin being spaced away from the device fins with gaps defined therein, and a third dielectric fin in the cut trench.
13. The method of claim 12, wherein, after the selectively removing of the upper portions of the first dielectric material layer,
  - the first dielectric layer wraps around bottom portions of the second dielectric layer; and
  - the second and third dielectric layers have a coplanar top surface.
14. The method of claim 12, wherein
  - the second trench is wider than the first trench;
  - the first dielectric fin includes the second dielectric material layer and is free of the third dielectric material layer;
  - each of the second and third dielectric fins includes both second and third dielectric material layer; and
  - the second and third dielectric material layers are different from each other in composition.

**15.** The method of claim **12**, further comprising forming a gate stack disposed on the device fins, and the first and second dielectric fins, wherein

the first and second device fins and the first and second dielectric fins are longitudinally oriented along a first direction;

the first and second device fins and the first and second dielectric fins are alternatively configured and are spaced away from each other along a second direction that is orthogonal to the first direction, defining hybrid gaps each spanning between one of the first and second device fins and one of the first and second dielectric fins;

the third dielectric fin is longitudinally oriented along the second direction; and

the forming of the gate stack includes forming portions of the gate stack inserted in the hybrid gaps.

**16.** The method of claim **12**, wherein

the forming of the second dielectric material layer includes performing an atomic layer deposition (ALD) process to deposit a high-k dielectric material in the first and second trenches and the cut trench; and

the forming of the third dielectric material layer includes depositing a carbon-containing material on the second dielectric material layer in the second trench and the cut trench, using one of spin-on coating and flowable chemical vapor deposition (FCVD) such that the second dielectric material layer laterally surrounds the third dielectric material layer and contacts sidewalls of the third dielectric material layer in the second and third dielectric fins.

**17.** A semiconductor structure, comprising:

a first device fin and a second device fin formed on a substrate and longitudinally oriented along a first direction;

a first dielectric fin and a second dielectric fin formed on the substrate, longitudinally oriented along the first direction, and spaced away from the first and second device fins;

a third dielectric fin contacting sidewalls of the first and second device fins and longitudinally oriented along a second direction being orthogonal to the first direction; and

a gate stack disposed on the first and second device fins and the first and second dielectric fins and spaced away from the third dielectric fin.

**18.** The semiconductor structure of **17**, wherein

the third dielectric fin contacts sidewalls of the first and second device fins; and

the first dielectric fin includes a first width and the second dielectric fin includes a second width greater than the first width.

**19.** The semiconductor structure of **17**, wherein

the gate stack is longitudinally oriented along the second direction; and

a first portion of the gate stack is inserted in a first gap between the first device fin and the first dielectric fin, and a second portion of the gate stack is inserted in a second gap between the second device fin and the first dielectric fin.

**20.** The semiconductor structure of claim **17**, wherein

the first dielectric fin includes a first dielectric material layer and is free of a second dielectric material layer;

each of the second and third dielectric fins includes both the first and second dielectric material layers;

the first and second dielectric material layers are different from each other in composition, wherein the first dielectric material layer includes one of a metal oxide, a metal nitride and a combination thereof, and wherein the second dielectric material layer includes silicon and carbon; and

a top surface of the first dielectric material layer and a top surface of the second dielectric material layer in the second dielectric fin are coplanar.

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